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Low-Wage Workers and the Enforceability of Noncompete Agreements

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Abstract. We exploit the 2008 Oregon ban on noncompete agreements (NCAs) for hourly-paid workers to provide the first evidence on the impact of NCAs on low-wage workers. We find that banning NCAs for hourly workers increased hourly wages by 2%–3% on average. Since only a subset of workers sign NCAs, scaling this estimate by the prevalence of NCA use in the hourly-paid population suggests that the effect on employees actually bound by NCAs may be as great as 14%–21%, though the true effect is likely lower due to labor market spillovers onto those not bound by NCAs. Whereas the positive wage effects are found across the age, education, and wage distributions, they are stronger for female workers and in occupations where NCAs are more common. The Oregon low-wage NCA ban also improved average occupational status in Oregon, raised job-to-job mobility, and increased the proportion of salaried workers without affecting hours worked.

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Keywords: low-wage workers • noncompete agreements • wages • mobility • labor market frictions

1. Introduction

A growing body of research in management and economics recognizes the importance of labor market frictions in determining both worker and firm outcomes (Acemoglu and Pischke 1999, Campbell et al. 2012, Mahoney and Qian 2013). One friction that has recently received significant attention is the use of noncompete agreements (NCAs), which prohibit departing employees from moving to new jobs within the industry for a period of time postemployment (U.S. Department of the Treasury 2016, White House 2016). A key reason that NCAs have received such attention, especially from policymakers (Beck 2019, Chile 2019), is because of their recent publicization in low-wage jobs, where the traditional rationales for their use (e.g., protecting trade secrets) are unlikely to apply—such as sandwich makers,¹ camp counselors, unpaid interns, and doggy day-care workers (Greenhouse 2014). Indeed, whereas NCAs are frequently assumed to occur only in high-wage jobs (Friedman 1991), we find that the *modal* worker bound by an NCA is paid by the hour, with median wages of \$14.² These facts raise an important question for practice and policy: What are the effects of NCAs on low-wage workers?

The literature on NCAs has yet to adequately address this question, both because it is focused almost exclusively on high-wage occupations, such as executives and high-tech workers, and because the estimates

for these groups are mixed (e.g., Stuart and Sorenson 2003, Fallick et al. 2006, Marx et al. 2009, Garmaise 2011, Conti 2014, Marx et al. 2015, Jeffers 2017, Kini et al. 2019, Balasubramanian et al. 2020). Although the freedom-to-contract and investment arguments often made in the context of high-wage workers may also apply to low-wage workers (Rubin and Shedd 1981), we document two differences between low-wage and high-wage workers that make it difficult to assess *ex ante* which theories and results from this literature will extend to the low-wage population. First, we show that low-wage workers are more fungible across industries—potentially rendering NCAs ineffective as within-industry mobility constraints. Second, we show that low-wage workers have less bargaining power in contracting around NCAs—suggesting that freedom-to-contract arguments may have limited explanatory power.

In light of the gap in the theoretical and empirical literature and the attendant policy interest, in this study we exploit a 2008 reform in Oregon that banned NCAs for hourly and low-wage workers³ to provide the first causal evidence on how banning NCAs affects low-wage workers. Using difference-in-difference and synthetic control designs with data from the Current Population Survey (CPS), we find that the Oregon low-wage NCA ban increased hourly wages by 2.2%–3.1% on average, with effects as great as 6% over a seven-year period. If this whole effect is driven

by the 14% of hourly-paid workers estimated to be bound by NCAs, then the wage increase for those bound by NCAs may be as large as 14%–21%. However, the negative effects of NCAs may spill over to others not bound by them (Starr et al. 2019, Johnson et al. 2020), suggesting that the true effect for NCA-bound workers is between these two estimates.

These positive wage gains are found across the age, wage, and skill distributions. Moreover, whereas the wage effects are positive for men, they are approximately twice as large for women, suggesting that vulnerable workers who may be less likely to negotiate may be most affected by NCAs. We also leverage the fact that NCAs are more frequently found in some occupations (Starr et al. 2021) to perform a within-Oregon and triple-difference analysis that ensures that the results are not due to, for example, differential responses to the recession in Oregon.

NCAs are, of course, within-industry *mobility* restrictions, and so we bolster these results by showing that monthly job-to-job mobility among hourly workers increased by 12%–18% after the NCA ban, driven mostly by increases in within-industry mobility. We also find that the NCA ban improved the occupational status of Oregon workers by up to 4.5%, depending on the measure.

We study employer responses as well. We find no evidence that the NCA ban resulted in employers cutting hours worked per week, such that the observed hourly wage increases translated into increased *weekly earnings* of 2.2%–2.8% (as opposed to being offset by reductions in hours). Although hours are not cut after the NCA ban, firms could avoid the NCA ban by converting hourly-paid workers to salaried, as long as the worker earned more than the median household income of a family of four. Indeed, we find that, after NCAs were banned for hourly workers, workers in Oregon are more likely to be salaried, especially just above this earnings threshold recognized in the 2008 law.

These results are robust to a variety of potential control groups, treatment windows, control variables, synthetic control designs, and, for the most part, a new inference method designed to address difference-in-difference designs with few treated groups and heterogeneity in group size (Ferman and Pinto 2019). We also rule out several alternative explanations by analyzing trends in unemployment, gross domestic product (GDP) growth, housing prices, labor force demographics, and firm dynamics.

This study makes two important contributions. The first is to the policy debate and to the existing literature on NCAs. We show that, despite prominent arguments in the organizational literature emphasizing aligning incentives for investment (Rubin and Shedd 1981, Grossman and Hart 1986), when low-

wage NCAs are enforceable, any additional compensation low-wage workers receive due to quasi-rents generated by firm investment is overwhelmed by a decrease in compensation associated with the limited threat of within-industry mobility. This effect is also larger for women, suggesting that these mobility frictions may be particularly pernicious for traditionally vulnerable or disadvantaged workers. These findings are directly relevant to the dozens of current state and federal proposals to ban NCAs for low-wage workers (Beck 2019).

Second, whereas research in management and economics emphasizes how frictions can help secure rents from workers (Barney 1991, Campbell et al. 2012), this view has generally not been associated with low-wage workers—especially in the literature on NCAs and trade secrets—since it is typically assumed that low-wage labor markets are perfectly competitive (Stigler 1957). Our study challenges this view by highlighting that managers were indeed able to use NCAs with low-wage workers to lower wage and turnover costs, allowing the firm to capture more value from workers, as long as productivity did not fall too much. We thus join a growing body of literature that also finds that firms have labor market power in low-wage labor markets (Card and Krueger 1994, Cengiz et al. 2019, Dube et al. 2018).

2. Motivation and Theory

Historically, the literature on NCAs has focused on workers earning relatively high wages: for example, chief executive officers (Schwab and Thomas 2006, Garmaise 2011, Bishara et al. 2015), physicians (Lavetti et al. 2019), and engineers (Fallick et al. 2006, Marx et al. 2009, Marx 2011, Marx et al. 2015, Balasubramanian et al. 2020). A few recent studies, however, have brought to light their prevalence economy-wide. Starr et al. (2021) find that 18.1% of the U.S. labor force in 2014 were bound by an NCA, including 13% of those earning less than \$40,000 annually, and Johnson and Lipsitz (2020) similarly find that 30% of hair stylists signed NCAs. Despite the fact that NCAs are more prevalent among high-wage workers, hourly workers make up 53% of NCA signers across the United States (based on data from Starr et al. 2021) because they make up such a large segment of the U.S. labor force.

Although our divergence from the prior literature is motivated partially by the empirical prevalence of low-wage NCAs, we show that there are two distinctions between low-wage and high-wage workers that may underlie fundamentally different labor market outcomes. The first distinction is that low-wage workers are more likely to transition across industries, which may cause NCAs to have limited consequences if moving jobs is frictionless (Neal 1995,

Mayer et al. 2012, Sykuta 2014). Figure 1 establishes this fact using data from the March supplement to the CPS: it shows that the likelihood of switching industries is much higher for low-earning workers by correlating industry changes with earnings in the individual's longest held job of the prior year.

The second distinction is that low-wage workers may have less leverage in the NCA signing process, perhaps due to the cost of hiring legal assistance. Table 1 uses data from Starr et al. (2021) to establish these differences by examining how several characteristics of the NCA contracting process differ between salaried and hourly workers using nationally representative data. Relative to salaried workers, NCA-bound hourly workers are more likely to sign their NCA without reading (8.8% versus 4.3%) and are substantially less likely to consult with friends or family (6.7% versus 14.5%) and with lawyers (2.8% versus 11.7%). Hourly workers are also much less likely to report negotiating over their NCA (4.5% versus 14.8%). Furthermore, hourly workers report that they are less likely to be promised anything in exchange for signing an NCA (90% versus 83.6%). Taken together, Table 1 shows that low-wage workers are clearly in a less advantageous bargaining position when it comes to signing an NCA.

In the remainder of this section, we draw on these distinctions to consider the theoretical relationship between low-wage NCAs and labor market outcomes.

Ex ante, it is unclear how NCAs will affect low-wage workers and firms' ability to extract rents from them. Accordingly, we explore possible outcomes and the assumptions that underlie each one.

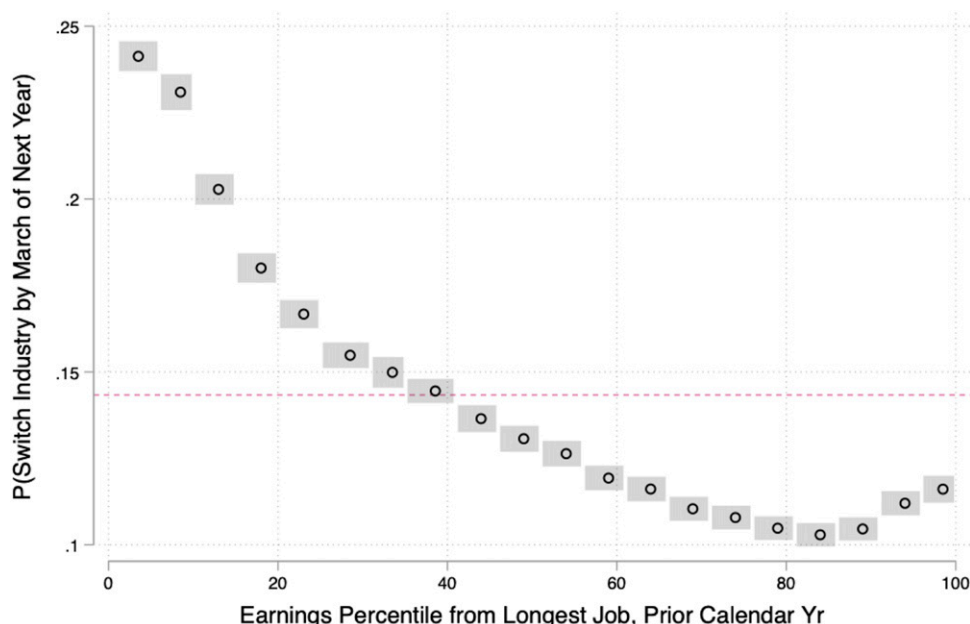
2.1. Why Might NCAs Be Unrelated to Wages?

Low-wage labor markets are often viewed as employing workers who are relatively substitutable across industries or require little specialized training (Stigler 1957, Sykuta 2014). If low-wage workers are highly substitutable across industries, as Figure 1 suggests, then it will be difficult for firms to extract rents under an NCA (which imposes costs only on *within-industry* mobility). When a firm attempts to keep a worker's wages low by threatening to enforce an NCA, that worker may simply look for work in another industry, which will pay just as much if the worker's human capital is perfectly fungible across industries.

2.2. Why Might NCAs Increase Wages?

Organizational scholars recognize the value of NCAs as a resolution to a holdup problem for the firm (Williamson 1975). Absent an NCA, if the firm invests in a costly but beneficial asset to production (i.e., industry-specific training, passing on a trade secret, or acquiring clients), a worker may prevent the firm from capturing the return on their investment by joining or starting a competitor, or demanding a higher salary by threatening to depart. As a result,

Figure 1. Industry Switching Behavior by Earnings Percentile



Notes. Data from 1988 to 2016 from the March CPS. The dependent variable is an indicator equal to one if the industry in the prior calendar year for the longest job is not the same as the industry of the worker's main job in March of the next year. Earnings percentiles are calculated on a yearly basis, according to the workers earnings in the longest job in the last calendar year. The dotted line is the unconditional mean probability of leaving the industry: 0.14.

Table 1. Differences in the NCA Contracting Process by Compensation Structure

Panel A: What did you do when you were asked to sign a noncompete?		
	Hourly	Salaried
Signed without reading	8.8%	4.3%
Read quickly and signed	35.6%	29.3%
Read slowly and signed	52.6%	59.3%
Consulted with friends, family, and relatives	6.7%	14.5%
Consulted with a lawyer	2.8%	11.7%
Panel B: Did you negotiate over the terms of the noncompete or for other benefits in exchange for signing?		
	Hourly	Salaried
Negotiate over NCA	4.5%	14.8%
Panel C: What were you promised by your employer for signing a noncompete?		
	Hourly	Salaried
Not promised any benefits	90.0%	83.6%
Promised increased compensation	5.8%	8.0%
Promised more training	4.4%	6.7%
Promised job security	3.8%	8.7%
Promised more responsibility	2.3%	6.1%
Promised better working conditions	2.7%	4.9%
Promised more trust	2.7%	8.2%
Promised a promotion	2.2%	4.8%
Promised more information	1.7%	5.2%
Promised more clients	1.2%	3.7%
Promised something else	0.7%	1.5%
Promised more referrals	0.0%	3.5%

Source. Authors' calculations with data from Starr et al. (2021).

the firm may be disincentivized from investing, leading to relatively low productivity employment relationships. The firm may contract around this problem, but, due to the inalienability of human capital, firms cannot write complete contracts that indenture the worker to the firm, or separate the worker from his or her knowledge (Grossman and Hart 1986, Hart and Moore 1990). Accordingly, NCAs—by preventing the movement of the worker, and thus the transfer of information—can create incentives for firms to invest in and allow workers access to valuable information or training (Rajan and Zingales 1998). The worker will likely agree to such restrictions as long as they share in the increased productivity.

Even without such investments, workers may earn higher wages with NCAs by negotiating for higher pay to compensate them for the lost option value of changing employers in the future. Workers who accurately assess the cost of that lost option value may require extra compensation for signing an NCA, only agreeing to one if they are better off relative to their outside option.

Taken together, the theories that workers will share in rents and will require compensation for their lost option value—often justified based on an individual's "freedom-to-contract" (Blake 1960, Callahan 1985)—suggest that, under an NCA, workers would have higher earnings and lower mobility (because they would be better off in their current job).

2.3. Why Might NCAs Reduce Wages?

When workers are not highly substitutable across industries, and when workers do not fully understand their employment contract or have a limited ability to negotiate for higher pay to compensate them for lost (future) mobility, NCAs may reduce worker earnings while also limiting mobility. If workers are not highly substitutable across industries (or, equivalently, moving costs are large), then workers seeking to move to a firm outside the scope of the NCA will need to take a pay cut, since their industry-specific human capital will not transfer (or they will have to pay moving costs). Given the lack of options outside the industry, NCAs may therefore prevent workers from taking higher paying opportunities within the industry and will bind low-wage workers to firms that have no incentive to raise their pay, as threats to depart are not credible. As noted above, if workers understand this lost future value and demand a pay increase at the outset of employment, then their pay may still rise with an NCA.

However, workers may not properly estimate the costs that they bear from signing an NCA, or they may not be able to freely negotiate. Even "freedom-to-contract" advocates explicitly note that the assumptions of those models are unlikely to hold for low-wage workers (Callahan 1985, Friedman 1991).⁴ If, for example, NCAs are effectively invisible to workers (similar to so-called "no poach agreements"; see Krueger and Ashenfelter 2018 and Gibson 2019), then they may be unable to bargain for wage increases due to reduced future competition for their skills. This could happen if low-wage workers are unlikely to read their contracts (Ayres and Schwartz 2014, Krueger and Posner 2018) or if NCAs are provided after the worker has accepted the job when worker bargaining power is lower (Arnow-Richman 2006, Marx 2011, Starr et al. 2021). Low-wage workers may also underestimate the potential future harm done to them by the NCA (Koszegi 2014) and thus demand additional compensation that fails to cover the true harm. Regardless of the reason, NCAs will decrease workers' earnings on average by limiting their mobility if workers are not compensated for their lost future (potential) mobility.

3. Background on NCA Laws and Oregon's 2008 Policy

We discern between the aforementioned competing theories by exploiting a policy in Oregon in 2008 in which NCAs were banned for low-wage workers. In this section, we describe the details of that policy and the enforceability of NCAs generally.

In the United States, the enforceability of NCAs has traditionally been determined by state law (Malsberger et al. 2002), though some recent proposals have suggested federal policies (e.g., the MOVE Act) or regulation by the Federal Trade Commission (Open Markets Institute 2019). Only three states in the United States do not enforce NCAs for any workers (California, North Dakota, and Oklahoma), and these policies have been in place for some time (Arnow-Richman 2020). The remaining states enforce NCAs, though to varying degrees (Bishara 2011) based on their interpretation of a “reasonableness test,” which seeks to ensure that the employer is protecting a legitimate business interest with the NCA and that the worker and public are not unduly harmed by it. What is deemed reasonable, however, is the subject of debate across states, and several incremental changes in these laws have been exploited in the empirical literature (e.g., Garmaise 2011, Jeffers 2017, Starr 2019).

However, it is historically rare for states to make a wholesale change in their laws by voiding NCAs entirely or for some subset of the population (Beck 2019). In the last several years, as abuses of NCAs have gripped public attention and empirical work has documented some ill effects of their enforceability, dozens of low-wage bans have been proposed, with a few passing (Beck 2019). These changes are very recent, however, and thus difficult to study given the lack of data. Nevertheless, in a rare event for its time—and the first such ban of which we are aware—Oregon banned NCAs in 2008 for low-wage workers, and this is the policy shock we examine in this study.

Prior to 2008, Oregon enforced NCAs similarly to most states, using a set of “reasonableness” criteria meant to ensure that legitimate business interests were being protected without unduly harming workers or the public interest (Rassas 2008).⁵ On August 6, 2007, Oregon's governor signed Senate Bill 248, which made dramatic changes to Oregon's policy on NCAs, effective January 1, 2008, for new contracts (pre-existing contracts were governed by the old law). In contrast to the old law, the new law codified two specific provisions: first, it clearly protected low-wage and hourly workers by voiding their NCAs; and, second, it changed the conditions under which NCAs would be enforced. Along the first dimension, the bill specifically voids NCAs for everyone *except*

a worker “engaged in administrative, executive or professional work who: (a) Performs predominantly intellectual, managerial or creative tasks; (b) Exercises discretion and independent judgement; and (c) Earns a salary and is paid on a salary basis.”⁶ The bill also voids NCAs for individuals earning less than the “median family income for a four-person family.”

The motivation for the low-wage NCA ban echoes the same arguments that policymakers are making today.⁷ The Oregon Labor Commissioner at the time, Dan Gardner, justified the inclusion of these conditions by noting (without evidence) that “there is a dangerous expansion in the use of non-competition agreements in Oregon and Oregon workers are being unfairly prevented from working in their chosen fields of expertise” (Laws 2007). An attorney at the time also noted (without evidence) that NCAs were “used very widely in situations in which most legislatures [and] law professors . . . did not anticipate[,] . . . such as low-wage people in basically non-competitive industries[, and] in industries where changing jobs is common” and that NCAs were being used “as a threat to constrain people's ability to change jobs [and against] people that didn't have the sophistication to know that it might not be enforceable to them” (Rassas 2008, p. 456).

The second major addition to the Oregon NCA law made substantive changes to the conditions under which NCAs would be enforced. In particular, the law codified an early notice provision, deemed NCAs longer than two years unreasonable, clarified the types of business interests it considered necessary for enforceability, and, importantly, also required the employer to pay the employee during the prohibition period the greater of “(a) Compensation equal to at least 50% of the employee's annual gross base salary and commissions at the time of the employee's termination; or (b) Fifty percent of the median family income for a four-person family.” Other changes were geared toward on-air talent in the broadcasting industry (Rassas 2008, p. 447).

One of the key threats to a causal interpretation of results when using such policy shocks is that they may have been anticipated, so as to contaminate the trend in earnings in Oregon relative to control states. Fortunately, as described by Rassas (2008), who reviews the historical development of the Oregon law, there were three iterations of the law in the first year that it was introduced (2007), before it was signed into law, and only in the final iteration of the bill did it contain the conditions that voided NCAs for low-wage workers. Thus, the Oregon ban on NCAs for low-wage workers appears to have been surreptitiously added to the bill, suggesting that its adoption was at least unexpected at the time.

4. Data and Empirical Strategy

For the purposes of understanding the effects of banning NCAs on low-wage workers, as opposed to other changes in the law, we focus our empirical analysis on the subset of workers for whom NCAs were enforceable before 2008 but were clearly voidable after 2008. The policy identifies several types of workers for whom new NCAs would not be enforceable: workers who are paid hourly, workers who earn below the median family income for a four-person family, and workers who are salaried but do not work in managerial or professional work (as defined in the law). We focus on hourly workers for two reasons. First, it is the clearest identifier that an NCA is not enforceable, given that it is debatable which jobs fall under the umbrella of the overtime and minimum wage exemptions as referred to in the bill (U.S. Department of Labor 2016). Second, if we wish to focus on wages as an outcome, having the treatment group defined by a salary threshold creates untoward empirical problems, as in the minimum wage literature.

To examine how Oregon's NCA ban influenced the wages of hourly and low-wage workers, we need to know whether the worker is paid hourly, the worker's hourly wage, and ideally other relevant worker characteristics. Fortunately, the Current Population Survey's (CPS) Monthly Outgoing Rotation Group (MORG) data contain not only wage data but also information on whether the worker is paid hourly, along with several other characteristics. The CPS (Flood et al. 2018) interviews 60,000 individuals monthly for four consecutive months and revisits them eight months

later for another four consecutive months of interviews. At the end of each set of four consecutive monthly interviews (interviews 4 and 8 in the sample), the workers are asked about their wages (these respondents form the MORG). In these interviews, wage data are supplemented with questions about the worker's job and also include a question about monthly job-to-job mobility.

To adjust the CPS MORG sample to reach our main estimation sample, we first restrict the sample to those who work hourly. Second, we keep those who are in the private for-profit or private nonprofit sector and who are between the ages of 18 and 70. Third, we restrict the sample to five years before the ban (2003) and until 2014, when significant media scrutiny arose related to the national publicization of NCAs among low-wage workers. We revisit the sensitivity of our results to this window later. Summary statistics for variables of interest are shown in Table 2. It is notable that in Oregon approximately two-thirds of employed workers are paid by the hour, suggesting the NCA ban had an effect on the majority of workers. With the exception of the percentage of workers who are black, there are no differences that are particularly notable across the treatment and potential control groups (which we discuss shortly). We also note that nearly all hourly workers in our data are low-wage earners under the earnings threshold in the law—the median family income for a four-person family (Oregon Revised Statutes §653.295(1)(d)). The threshold in 2017 was \$79,626,⁸ whereas the 95th percentile of annual earnings (weekly earnings times 52) for hourly workers in our data was \$75,522.74. Hourly workers therefore

Table 2. Summary Statistics

	Oregon	Oregon census division	Oregon census region	Full United States
<i>Paid by the hour</i>	0.67 (0.47)	0.63 (0.48)	0.64 (0.48)	0.62 (0.49)
<i>Hourly wage</i>	16.86 (9.31)	17.58 (10.70)	16.95 (9.99)	16.41 (9.39)
<i>Age</i>	38.69 (13.24)	37.88 (13.01)	37.89 (13.17)	38.82 (13.44)
<i>Hours worked</i>	34.83 (11.57)	35.73 (11.13)	35.96 (11.36)	35.85 (11.30)
<i>Female</i>	0.50 (0.50)	0.48 (0.50)	0.49 (0.50)	0.51 (0.50)
<i>Black</i>	0.01 (0.12)	0.04 (0.20)	0.04 (0.19)	0.10 (0.30)
<i>No high school diploma</i>	0.12 (0.32)	0.17 (0.37)	0.15 (0.36)	0.13 (0.33)
<i>Covered by union</i>	0.11 (0.32)	0.14 (0.35)	0.11 (0.32)	0.10 (0.30)

Notes. Standard deviation in parentheses. All variables except *Paid by the hour* reflect the sample of hourly-paid workers, which comes from the full sample of employed workers aged 18–70 who are in the private for-profit or nonprofit sectors.

provide a clean “low-wage” treatment group for our purposes.

Our primary regressions estimate a difference-in-difference model of the following form:

$$Y_{ist} = \gamma_s + \lambda_{tr} + \beta X_i + \delta Post_t \times Oregon_s + \varepsilon_{ist} \quad (1)$$

where Y_{ist} is either log hourly wage, log weekly earnings (both of which are deflated using the Consumer Price Index, even though undeflated measures find very similar results), an indicator for being salaried, a dummy variable for changing employers in the prior month, or measures of occupational status; γ_s is a state fixed effect; λ_{tr} represents a year-by-month-by-census region fixed effect; X_i is a vector of individual level controls (age, age squared, union membership, union coverage, sex, race, and an indicator for education less than a high school diploma); $Post_t$ is an indicator for 2008 or later; and $Oregon_s$ is an indicator for the state of Oregon. We use the CPS MORG sampling weights in analyses that use variables only available in the MORG, and we use individual-level CPS weights elsewhere (unweighted results are markedly similar).

The coefficient of interest is δ , which estimates the impact of the policy by comparing the change in observed outcomes pre-2008 to post-2008 for the “treated” group compared with a “control” group. We focus most of our analysis on hourly workers, for whom NCAs were banned in Oregon in 2008, such that hourly workers in Oregon form the treatment group. Selection of the control group is less obvious. In theory, we wish to select as a control group those workers whose trend through the sample period most resembles what Oregon’s trend would have been absent the policy change. As this information is unobservable, numerous options are possible.

One natural approach is to select hourly workers in a state economically similar to Oregon. This is the approach used in, for example, Card and Krueger’s (1994) seminal analysis of the employment effect of the minimum wage. We slightly extend this methodology by using the full United States with geographically defined region-by-time fixed effects, such that our estimates are identified by comparing Oregon to its regional neighbors (wherein the states outside of Oregon’s region give us power by helping to identify the other coefficients). Our preferred specification is to use the census region as the geographic unit because in our synthetic control analysis these states comprise over 60% of the weight in matching Oregon’s pretrends. However, given the ad hoc nature of this choice, we report in the online appendix all of our analyses with three other controls groups: just Oregon’s census division (Alaska, California, Hawaii, and Washington), just Oregon’s

census region (Oregon’s census division, as well as Arizona, Colorado, Idaho, Montana, Nevada, New Mexico, Utah, and Wyoming), and the full United States with census division by time fixed effects. We also report synthetic control analyses in the appendix.

Another natural approach is to consider a comparison between workers who are more and less likely to be affected by the law. We exploit differences in the occupation-specific probability of NCA use (from Starr et al. 2021) to generate a within-Oregon comparison, which rules out concerns that some Oregon-specific shock is driving our results. Whereas it is also possible to compare hourly and salaried workers, this comparison is not meaningful since salaried workers are also treated by the significant changes in the 2008 Oregon law; we discuss this further in Section 5.1.

Finally, recent studies have discussed issues with inference in difference-in-difference research designs. Following Abadie et al. (2017), we report clustered standard errors at the level at which treatment is assigned, which has become common practice among applied researchers. However, prior research has also documented challenges to inference when there is only one treated group, as in this study, and there is no perfect method to do inference in this situation. Accordingly, we also report p -values from a placebo test using synthetic control methods, and in all of our main results we calculate p -values using the method in Ferman and Pinto (2019), which is specifically designed for few treated clusters. However, as Ferman and Pinto (2019, p. 453) note, “No heteroskedasticity-robust inference method in DID performs well with one treated group.”

5. Wage Results

In this section, we employ a difference-in-difference methodology to estimate the impact of the Oregon ban on NCAs for hourly-paid workers.

5.1. Baseline Results

Table 3 contains the results from our baseline difference-in-difference model on hourly wages. The first three columns report coefficients from difference-in-difference models that compare changes in Oregon after the ban with changes in other states. The coefficients may be approximately interpreted as semi-elasticities: an estimate of 0.025 implies that the impact of implementing the NCA ban was a ($e^{0.025} - 1 =$) 2.5% increase in hourly wages for workers in the sample. Column 1 reports the uncontrolled 2×2 difference-in-difference (i.e., the model expressed in Equation (1) with no controls, and no time or state fixed effects, but with dummy variables for post-2008 and Oregon); column 2 includes census region by year-month fixed

Table 3. Baseline Difference-in-Differences Estimates for Hourly Wages

Dependent variable: Ln(hourly wage)	Full United States, with census region-time fixed effect			Within high-NCA-use occupations, across states (4)	Within-Oregon, high- versus low-use occupations (5)	Triple difference	
	(1)	(2)	(3)			(6)	(7)
<i>Post2008</i>	−0.022*** (0.003)						
<i>Oregon</i>	0.022* (0.011)						
<i>Post × Oregon</i>	0.031*** (0.003)	0.025*** (0.004)	0.023*** (0.004)	0.045*** (0.005)		0.012*** (0.004)	
<i>High-use occ.</i>					0.232*** (0.048)		
<i>Post × High-use occ.</i>					0.045** (0.019)		
<i>Post × Oregon × High-use occ.</i>						0.033*** (0.004)	0.032*** (0.003)
Observations	991,219	991,219	991,219	300,520	14,817	991,219	991,219
R^2	0.001	0.016	0.222	0.204	0.310	0.281	0.287
State fixed effect		✓	✓	✓			
Year-month-region fixed effect		✓	✓	✓			
Individual controls			✓	✓		✓	✓
Year-month fixed effect					✓		
Year-month-region-high-use fixed effect						✓	✓
State-high-use fixed effect						✓	✓
Year-month-state fixed effect							✓
Ferman and Pinto (2019) p -value	0.078	0.267	0.259	0.025	0.000	0.156	0.025

Notes. All samples include only hourly-paid workers. Columns 1–3 include the full United States, building to a model with year-month-region fixed effects, state fixed effects, and individual controls. Individual controls include age, age squared, union membership, union coverage, sex, less than high school education, and dummies for race. Column 4 replicates column 3, except that it includes only high-NCA-use occupations (defined as above-average NCA-use by Starr et al. 2021). Column 5 examines a within-Oregon model comparing high-NCA-use to low-NCA-use occupations, using individual controls and year-month fixed effects. Columns 6 and 7 examine a triple-difference model leveraging heterogeneity across time, states, and within states between high- and low-use occupations. Column 7 subsumes the *Post × Oregon* coefficient due to the inclusion of year-month-state fixed effects. Standard errors clustered by state are in parentheses in every column except for 5, where they are clustered by occupation.

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

effects with separate state fixed effects (as in Equation (1)); and column 3 adds individual-level controls.

Overall, the estimates suggest that wages for individuals paid hourly increased by 2.3%–3.1% following implementation of the NCA ban. Inclusion of individual-level controls in column 3 does not affect the estimates meaningfully, suggesting that demographic changes in the population of Oregon compared with control states did not drive the estimated impact of the NCA ban.

The primary concern with the estimates reported in columns 1–3 is that, instead of identifying a causal treatment effect of the NCA ban, they may reflect Oregon-specific shocks, such as a differential response to the Great Recession. We show that this is not the case by exploiting variation in the rates of NCA use by occupation (Starr et al. 2021), which allows us to examine within-Oregon models and triple interaction models that include state by month fixed effects. To begin, we construct an indicator variable that

takes the value of 1 if NCA use in the observation's occupational class (using two-digit Standard Occupational Classification codes) was above the national rate of use (18%). The model in column 4 is identical to the model in column 3, but is estimated only for observations in which the individual works in a high-NCA-use occupation. The effect of the NCA ban in occupations in which NCAs are used at a high rate was approximately double the effect observed in all occupations (4.5% versus 2.3%). In column 5, we compare high- and low-NCA-use occupations *within Oregon*: whereas the sample becomes significantly smaller due to the change in comparison group, variation in responses to the Great Recession across states (or even idiosyncratic shocks to Oregon that affect *all* hourly workers) have no ability to pollute the estimates. Again, we estimate that the NCA ban increased wages of workers in high-NCA-use occupations in Oregon by 4.5% relative to workers in low-NCA-use occupations in Oregon.

Though columns 4 and 5 address concerns over idiosyncratic state-level shocks (or idiosyncratic shocks to high-NCA-use occupations) that may confound our primary results, we take one further step by incorporating this variation in a triple-difference specification. In column 6, we augment the baseline model reported in column 3 with an interaction between the indicator for high-NCA-use occupations and $Post \times Oregon$ (as well as interactions of *High-use occ.* with the time and state fixed effects separately, allowing for time trends and state effects specific to high-versus low-use occupations). This specification identifies how the within-state difference between high-NCA-use and low-NCA-use occupations differs before and after the NCA ban in Oregon relative to the same difference in control states. The effect of interest is the coefficient on $Post \times Oregon \times High-use\ occ.$: there was a 3.3% greater impact of the NCA ban on workers in high-NCA-use occupations in Oregon, compared with the effect on workers in low-NCA-use occupations in Oregon (where both are being compared with out-of-state workers in similar NCA-use occupations). Finally, in column 7, we include state by year-month fixed effects, which subsume the $Post \times Oregon$ coefficient but allow the triple interaction term to be identified based on within state-month-year comparisons between occupations more likely and less likely to be bound by NCAs. The effects are nearly identical, suggesting that our main results are not driven by trends specific to Oregon as a whole, idiosyncratic state-level shocks that occurred concurrently with the change in NCA policy (e.g., a differential response to the Great Recession, which we explore further in Section 8), or even shocks that idiosyncratically affected *hourly* workers in Oregon post-2008.

Recent studies have identified issues in conducting inference in difference-in-difference studies, especially when the number of treated clusters is small or when there is heterogeneity in the number of observations in each cluster (e.g., Donald and Lang 2007, Conley and Taber 2011). Therefore, for each of our main results, we calculate p -values using the method of Ferman and Pinto (2019) (hereafter, “FP p -values”), which improves upon prior inference procedures when the aforementioned problems are present. We note that, whereas the p -values generated by this method are the current state-of-the-art, they are still inaccurate when there is just one treated group, as well as just 50 control groups (since asymptotics in FP are driven by the number of control groups going to infinity). We provide these estimated p -values as a guideline against which the standard cluster-robust p -values may be judged, but we admit to some degree of ignorance as to whether the true p -values are closer to the cluster-robust or FP p -values.

The FP p -values are reported for the coefficients of interest in the last line of Table 3 (adapting the procedure in columns 6 and 7 to account for the third difference). Although the FP p -values are not statistically significant at conventional levels across the board, we note that estimates in the specifications that one may expect, *ex ante*, to have the highest degree of precision (i.e., those leveraging variation in NCA use across occupations or focusing on occupations with the highest degree of use) are indeed highly statistically significant.⁹

The analysis contained in Table 3 estimates wage effects for the average worker, comparing groups of workers who were employed in Oregon and control states at different points in time. We may contrast this with a method using panel data, in which workers are tracked over time, and the relevant populations do not change. This is important for the interpretation of our results: in theory, it is possible that the change in wages is simply a composition effect. For example, workers sampled later in Oregon simply earned more than those sampled earlier, due to, for example, new workers moving to Oregon. To assuage this concern, we examine year-over-year wage growth for workers to see if rates of wage growth for individual workers changed with the NCA ban. This is made possible due to the short panel nature of the CPS: the MORG contains two interviews for most individuals, separated by one year. In Table OA1 in the online appendix, we estimate a difference-in-difference model using the log of wage growth over a one-year period for each worker. The results show that it was not just the level of wages that increased: within-individual wage growth went up by 1.3% following the NCA ban.

For full transparency, we replicate these results using a variety of control groups in Tables OA2 and OA3, showing that the choice of control group matters little for the results. Finally, we perform a triple-difference analysis using salaried versus hourly workers as a third difference, though we note that much of the bill passed in Oregon affected NCA use for salaried workers, which significantly pollutes the comparison: the results may be found in Table OA4 in the online appendix.¹⁰

5.2. Event Study Methodology

In order to interpret these coefficients as estimates of the causal impact of the NCA ban, it must be true that, absent the NCA ban, the trends in hourly wages in Oregon and control states would have evolved in parallel over time. Although we cannot test for parallel counterfactual trends in practice (since the counterfactual evolution of hourly wages in Oregon is unobserved), we may examine the trends in hourly wages in Oregon versus control states *prior* to

implementation of the ban as suggestive evidence that trends would or would not have evolved in a parallel fashion.

To do so, we examine both unconditional and conditional wage trends in Oregon relative to various control states. Figure 2 shows the unconditional time trends of average nominal hourly wages for hourly workers in Oregon as compared with various control states. In panels (a) and (b) of Figure 2, we observe that the wages of hourly workers in Oregon are approximately 20–60 cents below their counterparts, and these differences remain similar before 2008, as well as two years afterward. After 2010, however, hourly wages in Oregon rise relative to Oregon's neighbors and census region such that in 2012, workers in Oregon are earning approximately 80 cents more per hour. In panel (c), these trends are largely similar, except that relative to the full United States, there appears to be some evidence of a pretreatment increase in hourly wages in Oregon in 2007, suggesting perhaps that using the full United States as a control group (without adjusting for division- or region-specific trends, as we report in our main specifications) may not accurately reflect Oregon's postban counterfactual wages (were they not to ban NCAs).

We employ an event study methodology to examine these time trends while conditioning on several

control variables. The estimating equation is identical to Equation (1), except that we allow the Oregon effect to vary across years:

$$Y_{ist} = \gamma_s + \lambda_{tr} + \beta X_i + \sum_t \delta_t \text{Oregon}_s + \varepsilon_{ist}.$$

The event study coefficients are δ_t , which represent the marginal impact of living in Oregon at each year, t , relative to the omitted year. As is standard practice, we make the year before the ban (2007) the reference year. Given the difference-in-difference coefficients reported in Table 3 and the raw trends in Figure 2, we expect that the event study coefficients will be greater post-2008; however, a trend in the pretreatment period in the event study coefficients may suggest that our estimates are not identifying the causal impact of the treatment.

Results corresponding to our primary specification are presented in Figure 3, panel (a). For the full sample and the high-NCA-use subsample, there is little evidence of a pretrend, supporting the case that increases in earnings were due to the policy, not due to wages that were already growing in Oregon relative to control states (see Figure OA3 in the online appendix for other control groups). Panel (b) displays the results of a within-Oregon event study comparable to column 5 of Table 3. The graph plots the

Figure 2. Raw Trends in Hourly Wage by Control Group from 2003 to 2014

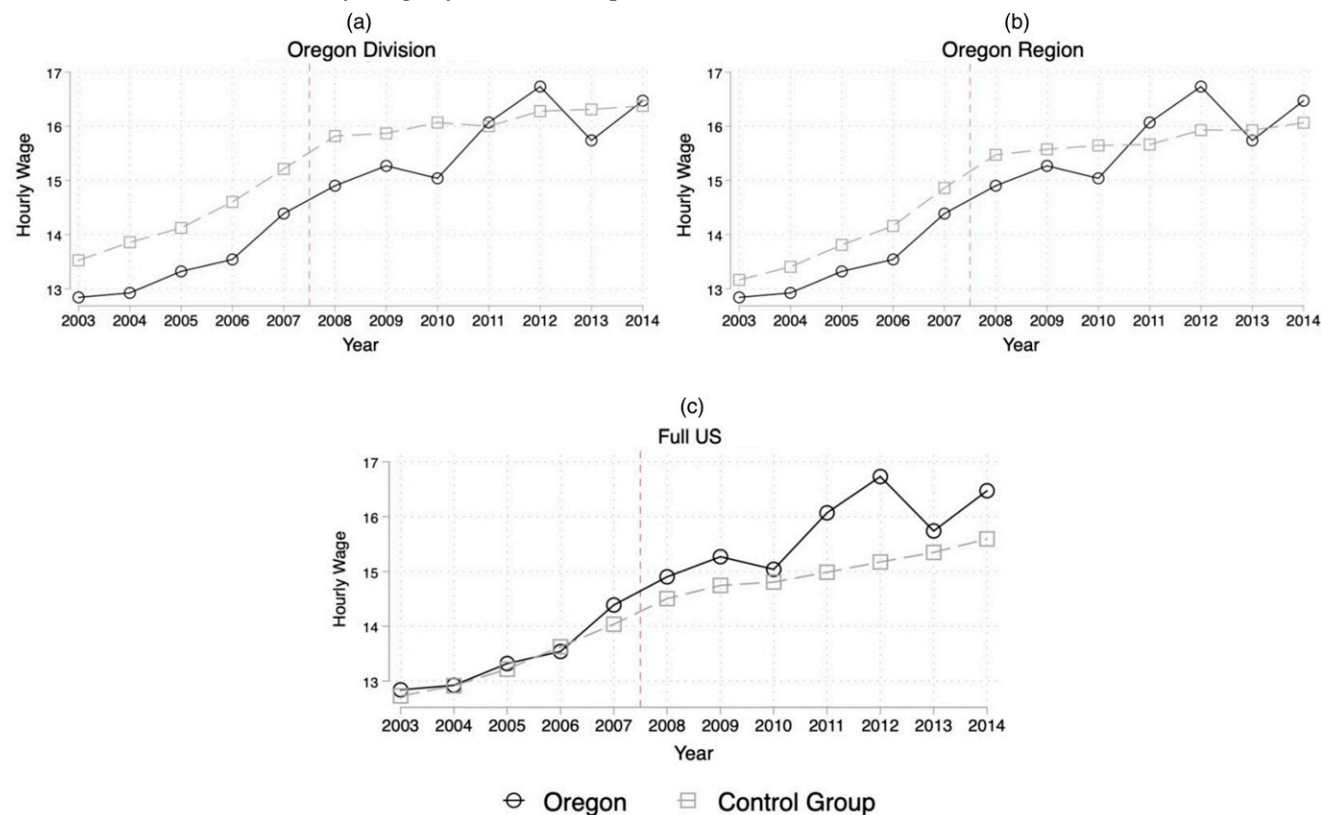
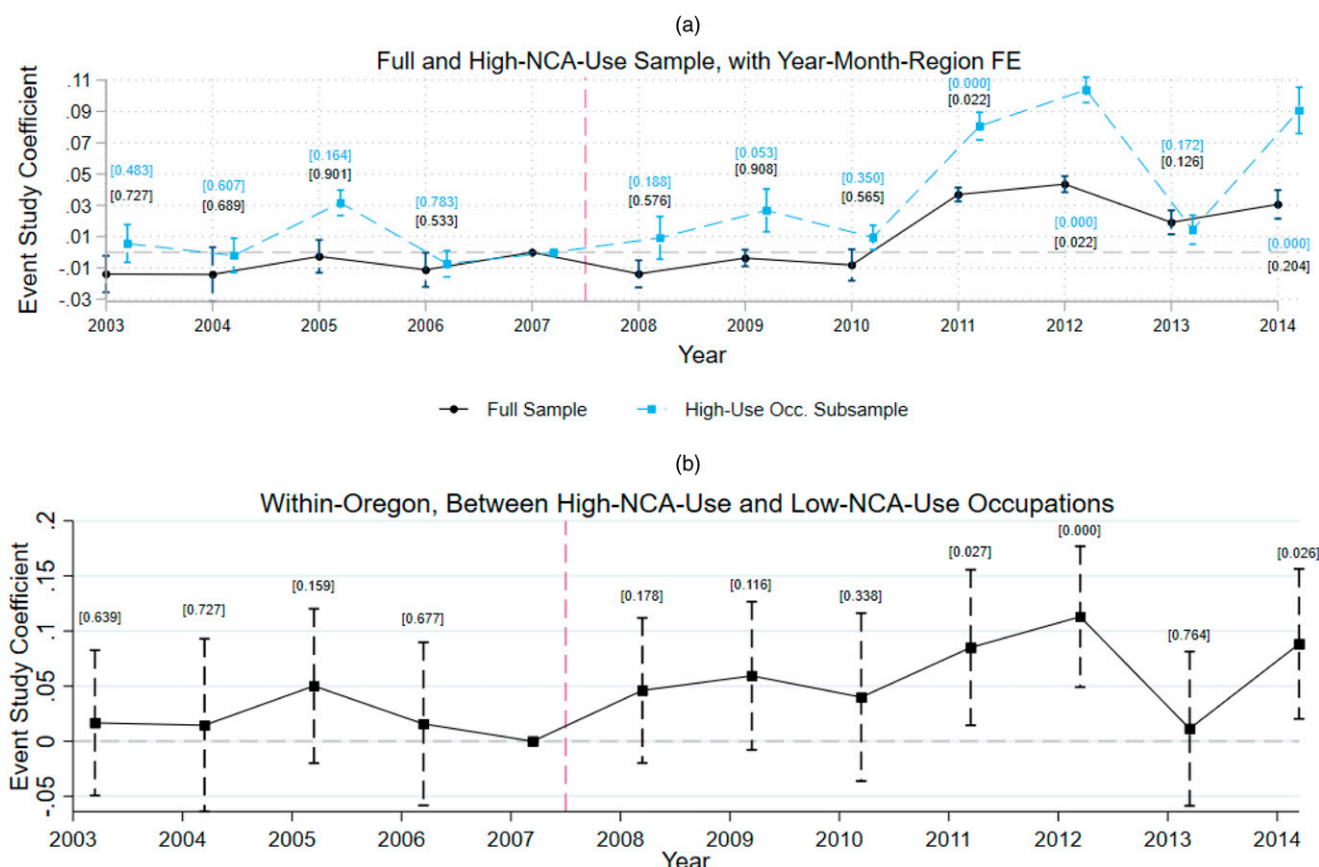


Figure 3. (Color online) Event Study Plots for $\text{Ln}(\text{Hourly Wage})$



Notes. The specification in panel (a) includes individual controls, in addition to year-month-region fixed effects, with standard errors clustered by state. The first model includes the full sample, and the second is the subsample of high-NCA-use occupations. The specification in panel (b) limits the sample to observations in Oregon and examines how hourly wages change between high-NCA-use occupations and low-NCA-use occupations. In this specification, standard errors are clustered by occupation. FP p -values are in brackets in panel (a), and wild bootstrap p -values are in brackets in panel (b).

estimated differences between high- and low-NCA-use occupations in Oregon over time. The results mirror those presented in panel (a). In both panels, FP p -values are in brackets, corresponding to each coefficient of each specifications. It is important to note that the p -values correspond to differences from zero, which here represents the omitted coefficient for 2007. To ensure that our choice of control group is robust to alternatives, in the appendix we show that our results are robust to synthetic control methods (with a p -value of 4% on the positive main effects for hourly workers).

We note that the slow increase in wages following the ban for the full sample is likely due to two factors: first, the ban did not nullify NCAs entered into prior to January 1, 2008. Therefore, any positive wage effects would likely be delayed, since in the years following 2008, Oregon workers whose tenure began before 2008 were still effectively subject to pre-2008 NCA law. Thus, as more workers entered new employment relationships over time (which was notably

rare over the course of the Great Recession) and thus became subject to the new law, we would expect the impact on wages to increase. Second, suppression of wages across the board due to the Great Recession may have limited wage growth in *all* jobs, preventing effects from appearing for a number of years. Reassuringly, the positive wage impacts are greater in the years immediately following the ban for workers in high-NCA-use occupations (both across states and within Oregon), suggesting that the ban had an immediate, albeit muted, impact.

5.3. Effects Across the Wage Distribution

Up to this point, we have examined the relationship between Oregon's NCA ban for low-wage workers and the *average* hourly wages of hourly-paid workers. In this section, we examine the effects across the *distribution* of hourly wages. If a ban on NCAs for low-wage workers is intended to help the most vulnerable workers (i.e., those at the bottom of the earnings distribution), then one concern is that positive *average*

effects are simply due to workers at the top of the hourly wage distribution earning still higher wages. Here, we show that this is not the case. To do this, we use the approach in Athey and Imbens (2006) to estimate treatment effects at quantiles of the wage distribution by comparing the treated distribution with a constructed counterfactual distribution. We present our results in Figure 4 (see Figure OA4 in the online appendix for other control groups), which shows that the positive average wage effects are positive across the whole range of the distribution, though the effects are stronger at the top.

5.4. Heterogeneity by Worker Characteristics

In this subsection, we explore heterogeneity in the wage effects of the low-wage NCA ban by several worker characteristics, including age, education, gender, occupation, and industry. These analyses are motivated by theoretical and practical concerns. First, we present estimates by age group. The motivation for this analysis is to test the hypothesis that workers beginning new jobs (e.g., young workers, who are by definition early in their career) were properly compensated for signing an NCA prior to the ban. If young workers are exempted from the wage increases present in the rest of the population, then this would be evidence that up-front compensation for signing an NCA is present for the workers most likely to be beginning new jobs. Next, we present estimates by education level. A natural question is whether the effect of the NCA ban scales with education. On one hand, those with more education are more likely to be bound by an

NCA (Starr et al. 2021), and so we should expect larger effects, though such workers may also be more likely to negotiate such provisions or otherwise mitigate the downsides of NCAs. With regard to gender, prior work has found that the enforceability of NCAs disproportionately discourages women from founding new firms following the dissolution of a prior firm (Marx 2018), and concurrent work finds that incremental changes in NCA enforceability have more severe effects for women (Johnson et al. 2020). Thus, one may wonder whether women also bear the brunt of the negative wage effects of NCAs in low-wage labor markets.

For each subsample of interest, we limit our data to that subsample and rerun our baseline wage specification with and without controls, presenting the $Post \times Oregon$ coefficient and 95% confidence interval, as well as the FP p -value in brackets for the estimate with controls. We note that inference is further complicated in these subgroup analyses by the limited number of treated observations included in each regression: for example, there are 842 observations in the postperiod in Oregon with less than high school education. Figure 5 depicts estimates of the treatment effects by age group, level of education, and gender. Notably, young workers (under age 25) experience a positive effect of the NCA ban that is similar in magnitude to the effect for workers ages 25–40. That is, even in the group most likely to be beginning new jobs, there is no evidence of a wage decrease, which would indicate that workers were being properly compensated for signing NCAs at the outset of a new

Figure 4. (Color online) Quantile Treatment Effect on Wages from Athey and Imbens (2006)

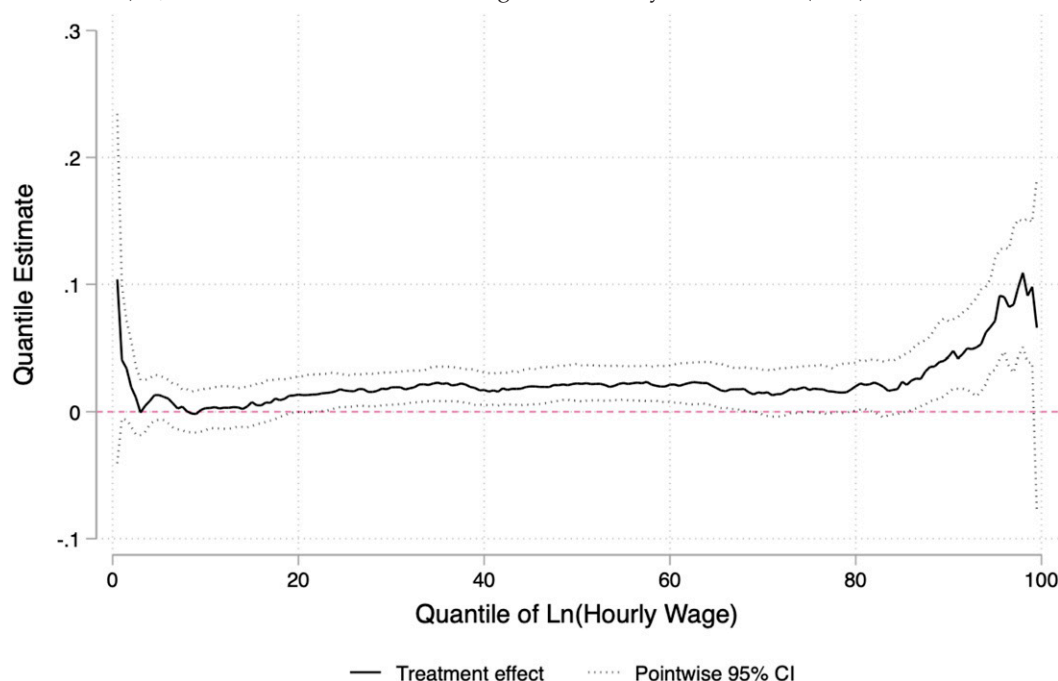
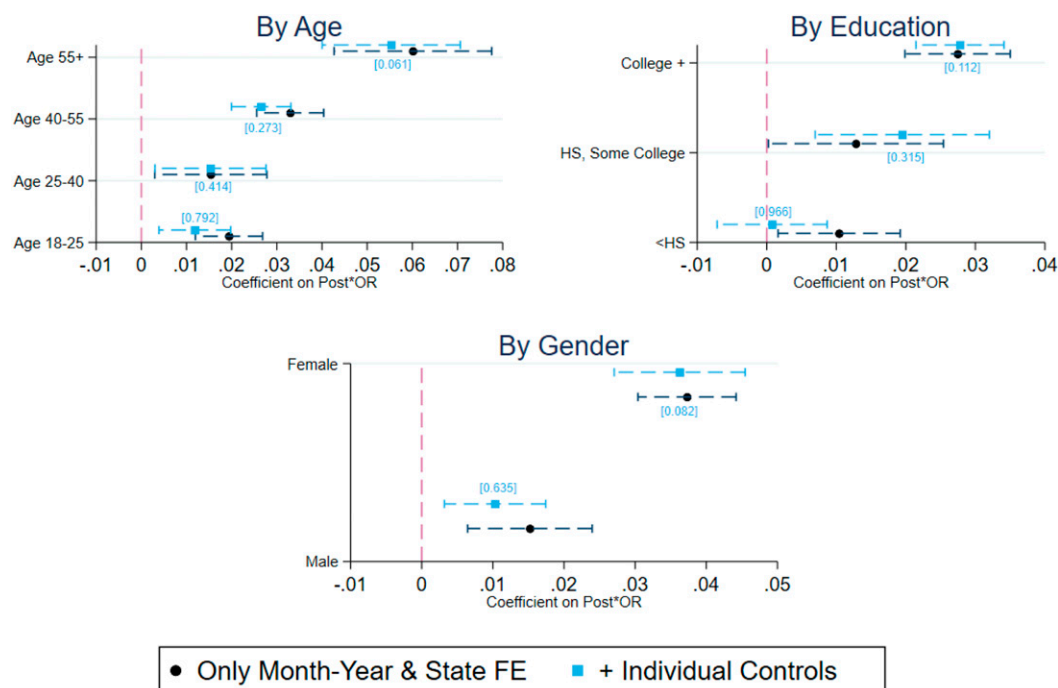


Figure 5. (Color online) Heterogeneity by Age, Education, and Gender



Notes. Each estimate is created by limiting the sample to each group and then estimating the main difference-in-difference model. That specification includes individual controls, in addition to year-month-region fixed effects and separate state fixed effects, with standard errors clustered by state. FP p -values are in brackets for the specification with controls.

job before the ban. We also note that the observed larger wage effects for late-career workers supports the causal interpretation of our results: these are the workers who have had the possibility of being bound by NCAs for the longest (leading to stunted wage growth), and therefore have the greatest potential to increase their wages after the ban.

Figure 5 also shows the effects across educational categories: we see that the positive wage effects following the ban are positive for both those with a college degree or higher and with a high school degree or some college (but no degree). For those without a high school degree, we find that the individual controls wash out an otherwise positive effect. Next, as in Johnson et al. (2020), we find that women disproportionately benefitted from making low-wage NCAs unenforceable, experiencing wage increases of 3.5% relative to 1.5% for men (see Figures OA5, OA6, and OA7 in the online appendix for other controls groups).

Finally, whereas we have documented that the wage effects are strongest in jobs that are more likely to use NCAs, one may wonder more generally about whether there are positive earnings effects across all occupations and industries. Figure 6 replicates Figure 5, splitting the sample separately by occupation and by industry. The positive wage effects are highest in management jobs (whose median wages

are \$17.7 per hour in this sample), installation, maintenance, repair, construction, and office support, but are persistently positive for almost all occupations, though to varying degrees depending on the control group. We see largely the same pattern across industries (see Figures OA8 and OA9 in the online appendix for other control groups). Importantly, in both graphs, it is clear that the effects observed in the baseline difference-in-difference specifications are not being driven solely by one or two industries or occupations. If that were the case, then changes in industrial or occupational composition of Oregon after implementation of the ban (especially changes related to the recession) may have driven the main effects. Since these effects are observed across occupational and industrial classifications, this is likely not the case.

6. Results on Occupational Status and Job-to-Job Mobility

In this section, we examine mobility and job quality outcomes. These results demonstrate that workers' abilities to move into better jobs increased after the low-wage NCA ban. The results are of independent interest, since worker mobility may improve worker-firm match-specific productivity (Stoyanov and Zubanov 2012, Jackson 2013) or worker well-being, and also serve to bolster the causal relationships identified above.

Figure 6. (Color online) Occupation and Industry Heterogeneity

Notes. Each estimate is created by limiting the sample to each occupation or industry and then estimating the main difference-in-difference model. That specification includes individual controls, in addition to year-month-region fixed effects and separate state fixed effects, with standard errors clustered by state. FP p -values are in brackets for the specification with controls.

6.1. Effects on Occupational Status

The analysis above examining wage effects *within* a given occupation may obscure the fact that the NCA ban may have allowed workers to access higher paying or more prestigious jobs. We test this hypothesis using a variety of measures of occupational status that consider prestige, income, and educational attainment. Given the debate over measures of occupational status, we use all such measures available from IPUMS USA (see Table OA5 in the online appendix for details). Using all observations from 2008, we calculate the average of each score using data from the American Community Survey (to which each score is linked on IPUMS.org) within the standardized occupational classification OCC1950 and merge the measures into our data using that variable in the CPS MORG. Because occupation alone may not capture all facets of a job, we supplement with the LASSO Industry Demographic and Occupation Score, or LIDO (Saavedra and Twinam 2020), which is specific to age-race-sex-industry-occupation cells.

Table 4 presents the results of difference-in-difference regressions in which the dependent variable is each

respective occupational status score, normalized to have mean zero and standard deviation one. For eight out of nine occupational status scores, there is a positive and statistically significant effect of the ban on occupational standing (though fewer are statistically significant when considering FP p -values). The positive effects range from 1.6% to 4.5% of a standard deviation of the relevant score. Although not huge, these effects suggest that there were increases in occupational status for low-wage workers in Oregon after the NCA ban.

6.2. Job-to-Job and Within-Industry Mobility Results

The positive impact of the ban on job quality does not necessarily indicate that workers engaged in actual interfirm mobility: it is possible that banning NCAs improved workers' abilities to bargain for better jobs within their own firm, improving their occupational title but not physically moving to a new firm. Here, we employ a nearly identical difference-in-difference regression in which we investigate the effect of the NCA ban on monthly mobility across employers.¹¹

Table 4. Difference-in-Difference Estimates of Occupational Status

Dependent variable	Duncan SEI	Hauser and Warren SEI	Siegel prestige	Nakao-Treas prestige	Occ score	Median earned income	% With one or more years of college	Nam-Powers-Boyd occupational status	LIDO score
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
<i>Post × Oregon</i>	0.035*** (0.008)	0.037*** (0.008)	0.037*** (0.008)	0.034*** (0.009)	−0.005 (0.004)	0.016*** (0.006)	0.045*** (0.011)	0.029*** (0.009)	0.016*** (0.005)
Observations	962,562	962,690	962,690	962,690	961,460	962,690	962,690	962,690	511,694
R ²	0.118	0.086	0.099	0.095	0.076	0.083	0.160	0.096	0.484
State fixed effect	✓	✓	✓	✓	✓	✓	✓	✓	✓
Year-month-region fixed effect	✓	✓	✓	✓	✓	✓	✓	✓	✓
Individual controls	✓	✓	✓	✓	✓	✓	✓	✓	✓
Ferman and Pinto (2019) <i>p</i> -value	0.102	0.127	0.039	0.080	0.988	0.475	0.083	0.194	0.256

Notes. SEI, socioeconomic index. Sample includes only hourly-paid workers. Sample sizes vary based on availability of the occupational standing score. For interpretability, all dependent variables are normalized to be mean zero, standard deviation of one. Standard errors clustered by state are in parentheses in each column. Details on each dependent variable are found in Table OA5 in the online appendix. See Duncan (1961), Hauser and Warren (1997), Siegel (1971), Nakao and Treas (1994), Nam and Boyd (2004), and Saavedra and Twinam (2020) for details on the measures of occupational status.

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

The baseline results using overall employee mobility are given in Table 5, column 1, and are reported with standard errors clustered at the state level, as well as FP p -values. The dependent variable is equal to one if the employee answered “No” when asked whether he or she works for the same employer as he or she did last month. The coefficient shows that the impact of the Oregon NCA ban was 0.40 percentage points. Against the average rate of monthly mobility (2.31% of workers), this coefficient corresponds to an increase in monthly mobility of approximately 17.3% of the sample mean. In column 2, we change the dependent variable to be an indicator equal to one only if the individual changed jobs *and* remained in the same census-coded industry. The rationale for this specification is that NCAs prohibit within-industry mobility (i.e., to competitors). Therefore, we expect the majority of the increase in overall mobility to be attributable to within-industry mobility, though there remains the possibility that spillover effects impact overall rates of job churn. The estimated coefficient from the within-industry specification corresponds to a 0.31-percentage-point increase in within-industry mobility, accounting for 79% of the overall mobility effect (Tables OA6 and OA7 in the online appendix show different control groups, whereas Table OA8 reports mobility results including those who are below the wage threshold).

Similar to our treatment of hourly wages, we examine trends in mobility prior to and following the enactment of the NCA ban. The unconditional, mean month-to-month job mobility trends for multiple

control groups are shown in Figure 7. Notwithstanding an uptick in mobility in 2005, Oregon appears to be on a downward trend relative to various control groups. This trend shifts, however, in 2008, when mobility in Oregon begins rising relative to the control states, such that in 2013 (at its peak), mobility rates in Oregon are actually above control states by approximately half a percentage point. In Figure 8, we depict both overall mobility and within-industry mobility, adjusting the raw mobility trends by conditioning on a variety of covariates using a similar event-study approach as in Figure 3, panel A (FP p -values in brackets). Whereas there remains noise in the preperiod in both series, there is no evidence of a pretreatment trend toward greater mobility (overall or within-industry) in Oregon relative to the control states (see Figure OA10 in the online appendix for various control groups). If anything, there is a slight downward trend in mobility in Oregon, relative to control states, prior to treatment. Nevertheless, after 2008, job-to-job mobility rises in Oregon and remains higher for the duration of the postperiod, both overall and within-industry, with the increases in within-industry mobility comprising nearly the full effect on overall mobility. Given the noise in the preperiod, we hesitate to interpret the coefficients as precisely estimated causal effects, though it is unequivocal that monthly mobility increased in Oregon relative to control states following the passage of the NCA ban.

Taken together, the observed occupation and industry changes show that occupation and industry fixed effects are “bad controls” in that the overall

Table 5. Mobility, Hours Worked, Weekly Earnings, and Changes in Compensation Structure

Dependent variable	1(Changed employer) (1)	1(Changed employer within-industry) (2)	Hours worked per week (3)	Ln(Weekly earnings)		1(Salaried) (6)	1(Salaried) (7)
				(4)	(5)		
<i>Post × Oregon</i>	0.400*** (0.048)	0.317*** (0.027)	0.029 (0.071)	0.024*** (0.007)	0.023*** (0.006)	0.014** (0.006)	−0.065*** (0.006)
<i>Ln(Hours)</i>					0.827*** (0.007)		
<i>Post × Oregon × above median income</i>							0.084*** (0.008)
Observations	944,184	928,423	991,554	988,305	988,305	1,610,190	145,634
R ²	0.006	0.002	0.116	0.219	0.448	0.100	0.121
Mean of dependent variable	2.31%	0.66%	35.86	6.228	6.228	39.0%	66.4%
State fixed effect	✓	✓	✓	✓	✓	✓	✓
Year-month-region fixed effect	✓	✓	✓	✓	✓	✓	✓
Individual controls	✓	✓	✓	✓	✓	✓	✓
Ferman and Pinto (2019) <i>p</i> -value	0.416	0.079	0.820	0.438	0.307	0.232	0.099
Sample	Hourly	Hourly	Hourly	Hourly	Hourly	All employed workers	All employed workers within \$10,000 of low-wage threshold

Notes. Coefficients in columns 1 and 2 are reported in percentage points. Standard errors clustered by state are in parentheses in every column.

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

effect of the NCA ban affects these choices as well. That is, if the ban itself caused changes in occupational and industrial composition (as shown in this section), inclusion of those fixed effects would misrepresent the overall impact of the ban. Table OA9 in the online appendix shows coefficients from our baseline regressions with the addition of occupation and industry fixed effects for each of our control groups: for example, the estimate in column 4, panel A, shows that about 65% (0.015/0.023) of the impact of the ban on wages occurred within occupations and industries, whereas the remaining 35% was due to changes in job composition in Oregon.

7. Results on Employer Responses

Employers subject to this NCA ban may respond on several potential margins, including cutting (or increasing) hours or changing the composition of the hourly and salaried workforce. We investigate these potential responses in this section.

7.1. Hours Worked and Weekly Earnings

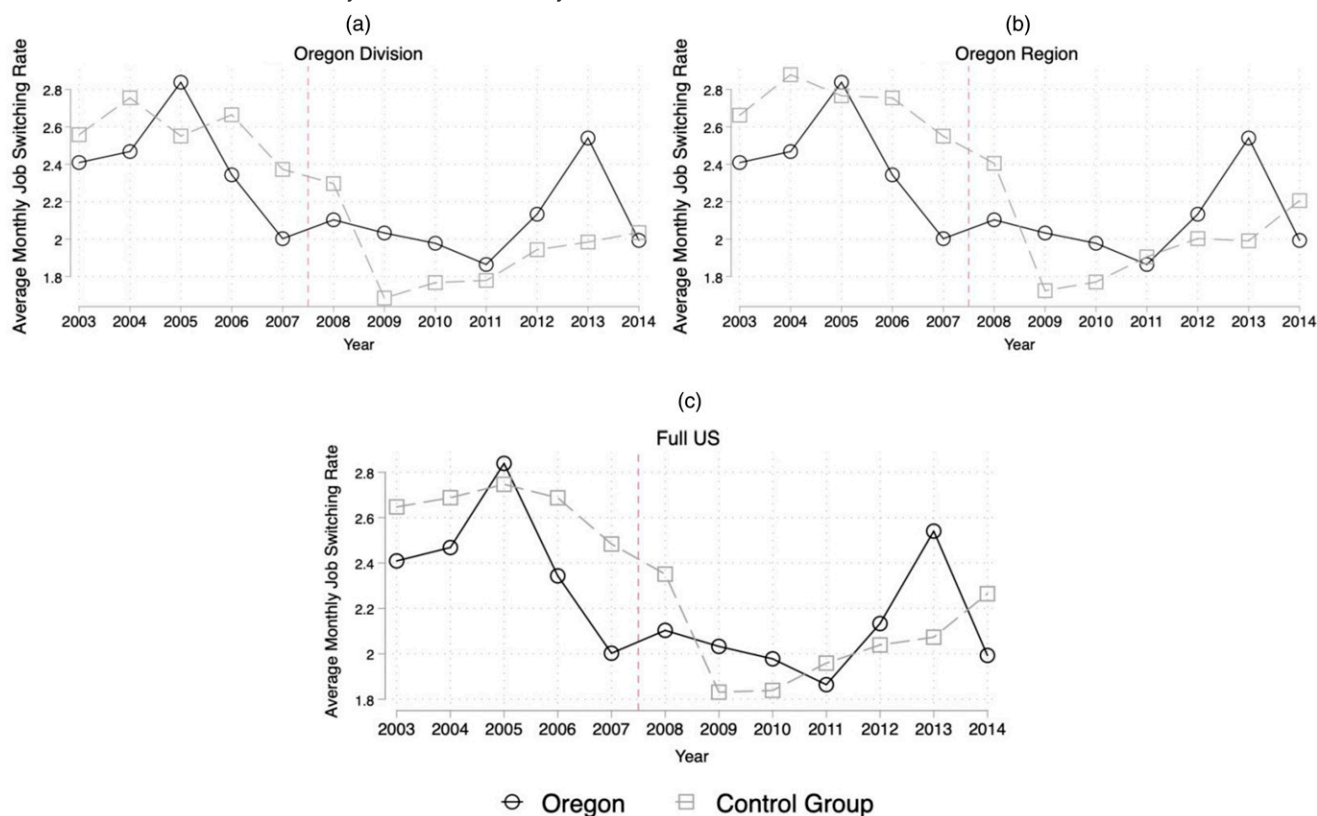
If wage increases are too large, then firms may cut worker hours, as some studies in the minimum wage literature have found (e.g., Stewart and Swaffield 2008). The result may be that our focus on hourly wages obscures effects on actual *earnings* if there are

commensurate cuts in hours worked. That is, if hours are reduced enough, then even though hourly wages rise, we may find mixed results on actual earnings. In Table 5, we repeat our main specification (Table OA10 in the online appendix reports results for various control groups), looking at the number of hours worked in column 3, and at log weekly earnings in columns 4 and 5, in which we examine models with and without controls for the log of hours worked. The results in column 3 suggest nearly no effect on hours worked: relative to control states, the average numbers of hours worked rose by about 0.02 hours per week—a rather small effect size of 0.05%. Given this small change to hours worked, it is perhaps unsurprising that, as shown in columns 4 and 5 of Table 5, the positive hourly wage effect carries over to weekly earnings, which rose 2.3%–2.4% following the ban.

7.2. Compensation Structure: Hourly vs. Salaried

One way that employers may avoid the effects of the NCA ban for certain employees would be to change their compensation structure from hourly to salaried. Assuming that the employee was not otherwise disqualified from the ban, an employer would still be able to use an enforceable NCA if it changed an employee's compensation from hourly to salaried. We test this idea by removing the sample restriction

Figure 7. Unconditional Monthly Job-to-Job Mobility Trends

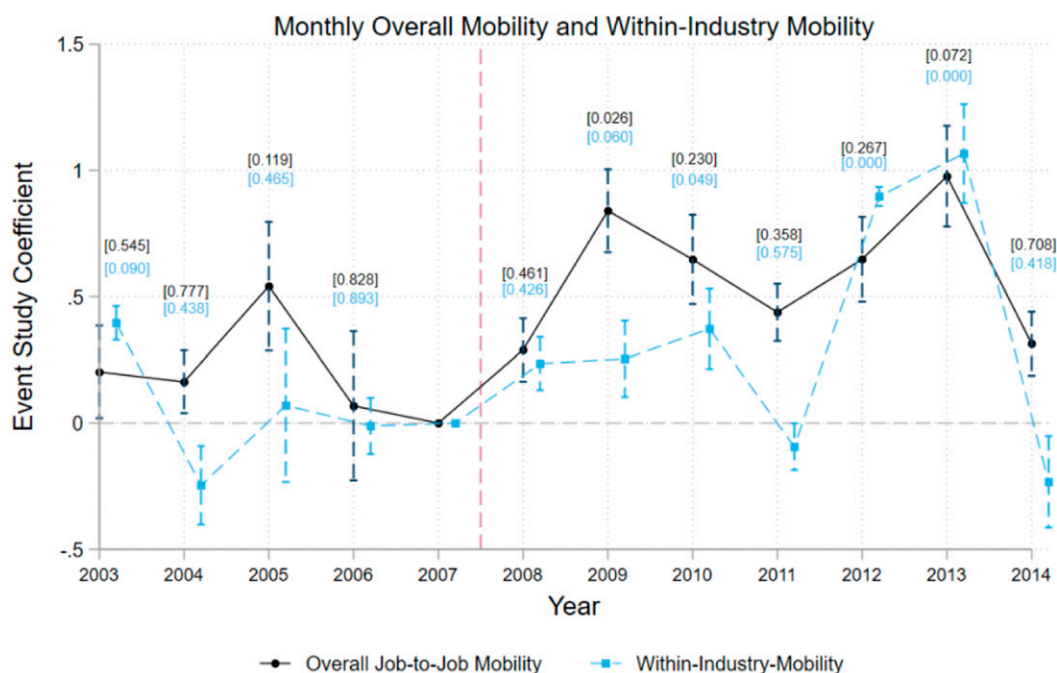


that employees are paid hourly from our main sample and run a difference-in-difference model with a dummy variable for whether the employee was paid a salary as the dependent variable. Baseline results are reported in Table 5, column 6. Relative to control states, the probability of a worker being salaried increased by 1.4 percentage points, compared with a sample mean of 39% in the full United States from 2003 to 2014 (a 3.6% increase).

The fact that the likelihood of being salaried rises following the NCA ban is suggestive evidence that employers are responding by giving workers more salaried jobs, though this may also be driven by the fact that workers are able to move into better jobs due to loosened restrictions on mobility (as in Table 4). Nevertheless, there is a clearer way to test the proposition that firms might respond to the hourly NCA ban by converting hourly workers to salaried, because the 2008 NCA law voided NCAs for salaried workers earning less than the median income for a family of four. In column 7 of Table 5, we restrict the sample to those earning \$10,000 above or below the median income cutoff and examine if those above the median income threshold are relatively more likely to be salaried after the NCA ban in Oregon relative to those just below the threshold (Table OA11 in the online appendix shows multiple control groups).

Whereas this procedure likely does not perfectly estimate the true causal impact of the ban on selection into salaried work (because earnings and therefore being above or below the cutoff depends on the policy), the results in column 7 suggest that indeed those earning just over the median income are 8.4 percentage points more likely to be salaried in Oregon following the ban relative to those whose earnings are just below the threshold (compared with control states).¹² In additional unreported results, we reran this analysis with \$5,000 and \$15,000 bins around the median threshold; we found very similar results for the \$5,000 bins and smaller but still statistically significant effects in the \$15,000 bins.

We view these results—especially those in column 7—as suggestive evidence that, rather than cutting hours, some employers manipulated job characteristics to avoid the NCA ban. This finding is important for three reasons. First, if salaried jobs are more desirable (due to greater stability, better benefits, etc.), then the NCA ban had the secondary effect of improving job quality, above and beyond the compensation and occupational status differences identified in the main analyses of this paper. Second, if workers' pay was switched to salaried instead of hourly after the ban, then those workers would not appear in the hourly sample in the postperiod of our main analysis. This would likely bias our results

Figure 8. (Color online) Event Study Plots for Monthly Mobility

Notes. This plot comes from an event study specification including the hourly-paid sample, with individual controls and month-year-region fixed effects and separate state fixed effects. The black line refers to a specification in which the dependent variable is an indicator for whether the individual changed employers in the last month. The blue dashed line corresponds to an identical specification, except that the dependent variable is an indicator equal to one if the individual changed employers and stayed in the same (census-coded) industry. FP p -values are in brackets with the top number corresponding to overall mobility and the bottom number corresponding to within-industry mobility.

downward, since these are the workers most likely to be paid at the top of the hourly wage distribution prior to the ban (as in Figure 4). Finally, this finding bolsters the causal interpretation of our results: the switching pattern suggests that employers were aware of the NCA ban and responded in meaningful ways that were not likely to be driven by other changes during the sample period, especially when combined with our other results.

8. Sensitivity Analyses

8.1. Treatment Window

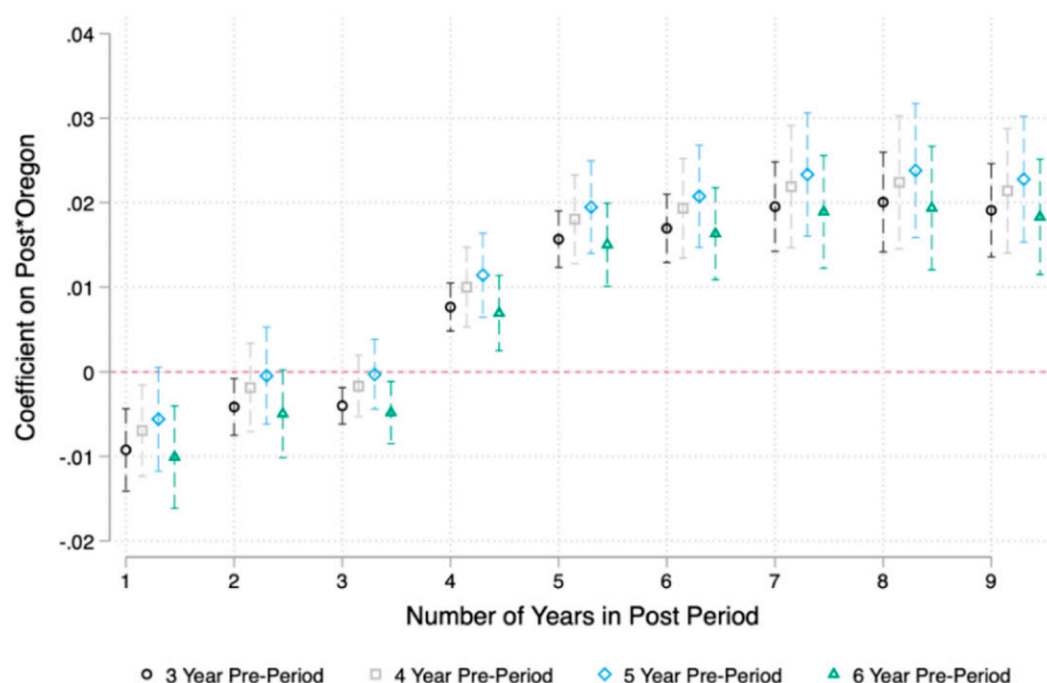
The baseline hourly wage estimates presented in Section 5 use data five years before and seven years after the policy change, from 2003 to 2014. We stop the analysis in 2014 because of the significant media attention following two high-profile cases of NCA use among low-wage workers,¹³ but, in this section, we ensure that our estimates are not driven by any particular window. In order to present our results with the greatest transparency possible, we show the coefficients arising from a wide array of pre- and postperiods in Figure 9 (Figure OA11 in the online appendix shows multiple control groups). Each marker on the graph is the coefficient from a separate regression (corresponding to column 3 in Table 2). Each series represents a different preperiod (from three

years to six years), and the horizontal axis indexes postperiods, from one year to nine years. Notably, the length of the preperiod does not play a large role in determining the effect of the NCA ban. The rise in the coefficients as the postperiod increases reflects the increased coverage (as fewer workers are covered by NCAs whose terms are governed by pre-2008 law). The coefficient stabilizes after approximately four to five years, remaining roughly constant to the end of the possible sample period.

8.2. Recession

The timing of the NCA ban in Oregon (January 2008) may confound our analyses if the mobility and wage effects of the recession are different for Oregon's low-wage workers. Whereas we use within-Oregon analysis in Sections 5.1 and 5.2 to demonstrate that a large variety of shocks or responses to the Great Recession do not explain our results, another approach is to simply control for the severity of the recession. We follow Giroud and Mueller (2017) in using housing value data from Zillow.com to proxy for the severity of the recession and include a control in Equation (1) for the natural logarithm of the median home value in the observation's county (as well as a fixed effect corresponding to the household's county). If the estimated effect of the NCA ban is due to a differential effect of

Figure 9. (Color online) Treatment Window Robustness for Wage Results



Notes. Each estimate is constructed by limiting the sample to the appropriate window defined by the graph and then running our main difference-in-difference model with individual controls and month-year-region fixed effects and separate state fixed effects.

the recession in Oregon counties versus control counties, then we would expect the coefficient of interest (reflecting the effect of the NCA ban) to shrink to zero. Reassuringly, this is not the case, as shown in Table OA12 in the online appendix.¹⁴

In addition, we examine whether Oregon's trends related to unemployment, GDP growth, housing prices, labor market demographics, and firm dynamics (entry, exit, changes in firm size) could explain our results. These analyses are reported in the online appendix, in Tables OA13–OA14 and Figures OA12–OA14. The results suggest that these trends are unlikely to explain the core patterns presented in this paper, either because the results are noisy, small in terms of magnitude, or because the temporal patterns do not align (i.e., although wages and mobility are rising, housing prices, GDP growth, and unemployment are flat).

8.3. Minimum Wage

Another possible concern is that other policies targeting low-wage workers, such as the minimum wage, may be exerting an undue influence on our results if policies in Oregon or control states coincidentally changed concurrent with the low-wage NCA ban. In Table OA15 in the online appendix, we control for state minimum wages and find little change in our results. Indeed, as the distributional analysis suggests, the positive wage effects occur throughout the

bottom half of the wage distribution. Nevertheless, other such policies in Oregon that influence the earnings of low-wage workers may bias our results. We reviewed the legislative history of Oregon between 2008 and 2014 and spoke to an economist at the Oregon Office of Economic Analysis, and we found no such changes (e.g., there were no changes to laws governing at-will employment or right to work laws).

8.4. Measurement Error and Weighting

Another concern with using wage data from the CPS, which is a household survey, is that measurement error in wages may inflate standard errors or otherwise bias our results if nonclassical, as in Bound and Krueger (1991). To partially address this concern, in Table OA16 in the online appendix, we reran our main analyses for hourly wages, weekly earnings, and mobility using only the sample of workers who themselves responded to the survey (as opposed to proxy respondents). As expected, by focusing on the workers who presumably know their exact wage and mobility patterns, the precision of the estimates increases, while coefficient directions are identical and the magnitudes are larger.

Finally, in unreported results, we reran our main specifications without the sampling weights, and the effects were largely unchanged in terms of magnitude and statistical significance.

9. Discussion and Conclusion

This study is motivated by the paucity of empirical evidence on the effects of NCAs among low-wage workers who likely have little access to valuable information, even though such workers actually comprise the majority of those bound by NCAs. We use CPS data to examine the effects of Oregon's 2008 NCA ban for hourly workers on hourly wages, occupational status, job-to-job mobility, and compensation structures. Our results suggest that, on average, banning NCAs increased the earnings of hourly workers by 2%–3%, with stronger effects for those in jobs most likely to sign NCAs, while raising monthly job-to-job mobility by 17%, moving workers into salaried compensation structures and into more prestigious and higher paying occupations. We also find that the positive wage effects accrue across the age, wage, and education distributions, and across multiple occupations and industries, though the effects are notably stronger for women versus men. These results are robust to a bevy of alternative specifications, control groups, treatment windows, and are modestly robust to alternative procedures for conducting inference.

How should we interpret this suite of results in light of the relevant literatures and theoretical arguments? Perhaps most directly, our results contradict the common arguments made in favor of enforcing NCAs. Under one theory, the lack of specialized skills among low-wage workers would render within-industry mobility restrictions ineffective. Under another, drawn from organizational theory and incomplete contracting, NCAs could benefit workers by resolving a holdup problem related to their on-the-job training or access to information. In contrast, our results show that Oregon's NCA ban for low-wage workers raised earnings and upward mobility for hourly workers, suggesting that these accounts of the low-wage labor market are inaccurate or incomplete.

One implication of these results for organizations is that firms can generally implement low-wage NCAs as a way to reduce labor and turnover costs. Whereas scholars in management and economics have long recognized the importance of labor market frictions like NCAs (Barney 1991, Campbell et al. 2012), this literature has focused almost entirely on high-skilled workers, such as inventors, engineers, and top management teams. Indeed, it is historically assumed that low-wage labor markets are perfectly competitive, such that firms are unable to extract rents from low-wage workers (Stigler 1957). Accordingly, our study recognizes that labor market frictions do matter for the firm's ability to lower turnover and wage costs for even low-wage workers.¹⁵ Extending beyond this study, these results imply that firms may also exploit NCA-induced power through nonwage means, such

that our estimates understate the full benefits of banning NCAs. They also have implications for the myriad other restrictive covenants that firms regularly use (e.g., confidentiality agreements, nonsolicitation agreements, forced arbitration agreements, and class-action waivers).

We cannot identify precisely where the “freedom-to-contract” argument falls apart, but we can identify some clues and mechanisms from our analyses, including some that are specific to the 2008 law. First, the fact that the 2008 Oregon law required “early notice” of the noncompete suggests that “late notice” noncompetes were frequent. Indeed, Starr et al. (2021) show that late notice of NCAs occurs approximately one-third of the time, and, when it does, those workers are worse off relative to those who have not signed an NCA. Combined with our finding that low-wage workers are severely disadvantaged in terms of their bargaining power over NCAs (see Table 1), delayed notification and low bargaining power suggest that low-wage workers simply accept mobility-constraining NCAs without demanding a compensating differential.

A second part of the 2008 law may have also played an important role in driving our results—the so-called “garden leave” provision that forced firms to pay workers during the prohibition period of the NCA. Prior literature has documented that NCAs are found frequently in states that do not enforce them (Bishara et al. 2015, Sanga 2018, Starr et al. 2021) and that even unenforceable NCAs have deterrent effects on employee mobility, reflecting so-called *in terrorem* effects (Blake 1960, Starr et al. 2020, Prescott and Starr 2021). This literature highlights that a policy in which NCAs are not enforced in court may not dissuade firms from using them, because even if the benefits from using NCAs are low, the costs may be low too. Thus the garden leave provision may have played an important role in reducing the actual use of (unenforceable) NCAs in Oregon after 2008 because it imposed a cost on employers for using them. Several recent bills have similar garden leave provisions (e.g., Massachusetts) or otherwise impose costs for using clearly voidable NCAs (e.g., the MOVE Act).

Finally, a key mechanism likely underlying the estimated wage gains are the job-switching patterns that we observe. Topel and Ward (1992) document that a large share of earnings growth comes from job switching. Whereas we cannot explicitly examine the wage effects for stayers versus leavers due to survey limitations of the CPS, the fact that mobility rises after the NCA ban suggests that there are wage increases for the leavers, though the stayers may also benefit from increased bargaining power in their current jobs. We hope that future work can address this shortcoming.

Several additional limitations of our analysis bear noting. First, and perhaps most important, we do not possess data on the actual use of NCAs. Our calculations from 2014 data suggest that 14% of hourly workers were bound by NCAs. If this is true, then a simple back-of-the-envelope calculation that assumes that the 2%–3% wage increases are driven only by 14% of the hourly population implies a treatment on the treated effect in the range of 14%–21%. However, because we do not possess data on the use of NCAs in our sample, we cannot actually determine whether the wage gains from banning NCAs for low-wage workers accrue only to those who were previously bound by NCAs or were shared by others in the market. Previous evidence documenting negative external effects of NCA use and enforceability suggests that an implied treatment effect on the treated of 14%–21% would overestimate the true effect. Similarly, we also cannot determine whether the use of NCAs fell after the ban took place, though due to the garden leave provision we think this is likely. To the best of our knowledge, there does not exist a panel data set with noncompete use for anything close to a representative sample of the United States, let alone for the low-wage labor force from 2003 to 2014. Such data gathering efforts are a clear avenue for future research.

Second, our analysis is based only on the policy changes of one state, which presents both empirical and inferential challenges. A key empirical challenge is that our results would be confounded by any unobservable event that affected the wages of Oregon's hourly population. We address this concern in multiple ways, including controlling for the recession, examining trends in recession-related characteristics, examining within-Oregon models, and bolstering a causal story by examining related outcomes like within-industry mobility, occupational status, and changes in the compensation structure around thresholds established in the law. Nevertheless, ideally, several states would have banned NCAs for low-wage workers so that we could separately analyze each ban or group them together to avoid such concerns.¹⁶ In terms of inference, there is no agreed-upon way to calculate *p*-values when there is one treated group. Accordingly, our approach is to report state-clustered standard errors, Ferman and Pinto (2019) *p*-values, and placebo tests in synthetic control models. Our main results across these approaches appear mostly robust, though in some specifications the discrepancy between these approaches is large.

Finally, our analysis focuses on worker outcomes and cannot directly address whether NCAs allow firms to actually capture rents from workers. Under the assumption that NCAs do not *decrease* workers'

productivity, our analysis provides evidence that firms are able to capture a greater share of rents by using NCAs, since wages decrease and turnover is limited. But it is theoretically possible that productivity under NCAs is also lower (e.g., if they discourage employee effort), although evidence on such effects is limited (Garmaise 2011, Lobel and Amir 2013, Starr 2019).

Limitations notwithstanding, our work provides novel evidence that—whatever reason firms use NCAs with low-wage workers—the effect is to consolidate labor market power and reduce labor and turnover costs. Accordingly, banning low-wage NCAs may help reverse the long-standing trends of wage stagnation for the lowest earning workers. Policymakers may also take note of the other relevant features of the law, as noted above, including how the garden leave provision likely reduced the use of NCAs by imposing a cost on firms (a cost that could be imposed by other means). The results herein, combined with the fact that more than 40 states do not currently have NCA bans for low-wage workers, make banning NCAs for low-wage workers an opportunity to improve the lives of the lowest earning workers.

Acknowledgments

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Appendix. Synthetic Control Analysis

The estimates presented in the main text are relatively robust across choice of control group, as shown in the tables and figures presented in the online appendix. However, given the nature of the policy change (occurring in just one state) and the possibility that control states may evolve in nonparallel fashion to Oregon, we employ a synthetic control methodology as a robustness check in this appendix. This methodology constructs a synthetic “state” by reweighting all possible control states such that the synthetic state most closely mirrors Oregon prior to the implementation of the ban. Therefore, the difference in pretreatment trends for Oregon versus the control group will be effectively minimized, assuaging any lingering concerns over pretreatment trends.

We implement the method pioneered by Abadie et al. (2010), which selects weights for each “donor” state (i.e., non-Oregon states) by minimizing the root-mean-squared prediction error (RMSPE) before treatment occurs. The weighted outcomes then form the control group against

Table A.1. Synthetic Control Weights

State	Weight	State	Weight
Alabama	0.4%	Missouri	0.7%
Alaska	6.6%	Montana	0.1%
Arizona	2.8%	Nebraska	0.4%
Arkansas	0.2%	Nevada	0.4%
California	0.8%	New Hampshire	0.6%
Colorado	25.5%	New Jersey	0.6%
Connecticut	2.4%	New Mexico	0.1%
Delaware	0.6%	New York	0.4%
District of Columbia	9.6%	North Carolina	0.3%
Florida	0.3%	North Dakota	0.4%
Georgia	0.4%	Ohio	0.4%
Hawaii	0.4%	Oklahoma	0.3%
Idaho	0.2%	Pennsylvania	0.8%
Illinois	0.5%	Rhode Island	0.3%
Indiana	1.1%	South Carolina	0.3%
Iowa	0.6%	South Dakota	0.2%
Kansas	0.5%	Tennessee	0.4%
Kentucky	0.3%	Texas	0.2%
Louisiana	0.3%	Utah	29.3%
Maine	0.4%	Vermont	0.4%
Maryland	1.0%	Virginia	0.5%
Massachusetts	0.6%	Washington	4.0%
Michigan	0.7%	West Virginia	0.1%
Minnesota	1.1%	Wisconsin	1.2%
Mississippi	0.3%	Wyoming	0.0%

which we can perform an event study analysis. To increase transparency and assuage common concerns associated with selection of pretreatment variables upon which to match, we follow Ferman et al. (2018) and conduct matching based only on outcomes in all pretreatment periods.¹⁷ The weights resulting from this matching procedure are reported in Table A.1. Notably, Colorado, Alaska, and Utah together (which are in Oregon's "census region" control group) receive over 60% of the weight, confirming that Oregon's census region is a reasonable choice of control group.

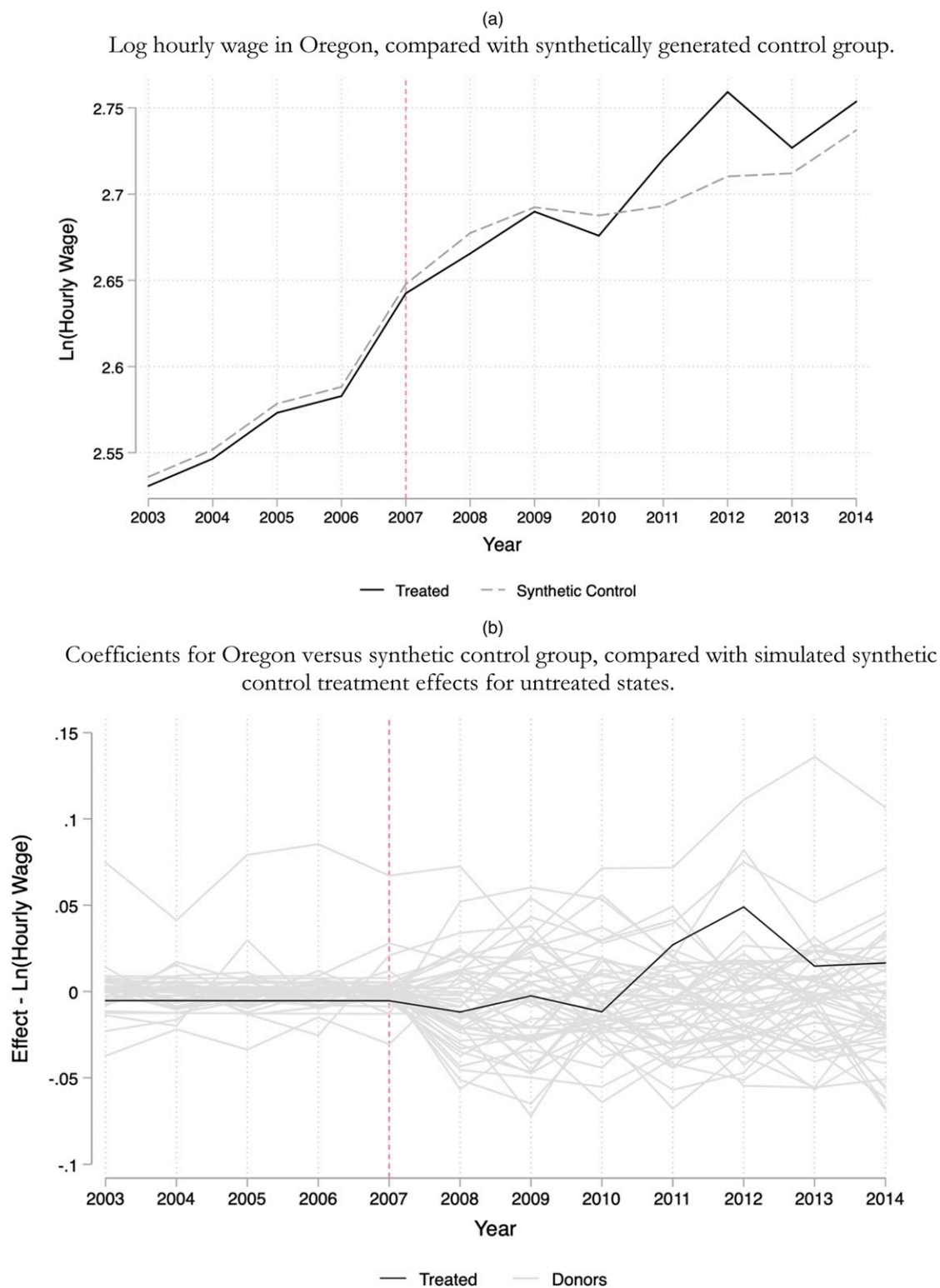
Panel (a) of Figure A.1 presents the synthetic control analogue of the unconditional wage trends presented for the three control groups in Figure 1. By design, the pre-2008 trends in log hourly wages (undeflated) are nearly identical, with divergence in the two series appearing after 2010, similar to the pattern using the other control groups. Panel (b) of Figure A.1 shows the analogue of the event study coefficients in the synthetic control framework.¹⁸ The dark series represents the event study coefficients for Oregon, compared with the synthetic control group that minimizes RMSPE in the pretreatment period for Oregon. The lighter series represent the same exercise, performed separately, as a placebo test, for each other state. That is, first, a synthetic control is generated for each other state that minimizes RMSPE in the pretreatment period for that state; then, event study coefficients are generated assuming that state was the treated state. Importantly, the event study coefficients in the posttreatment period for Oregon rise to the top of the distribution of coefficients for placebo states,

showing that the increase in hourly earnings for hourly workers in Oregon was large and positive.

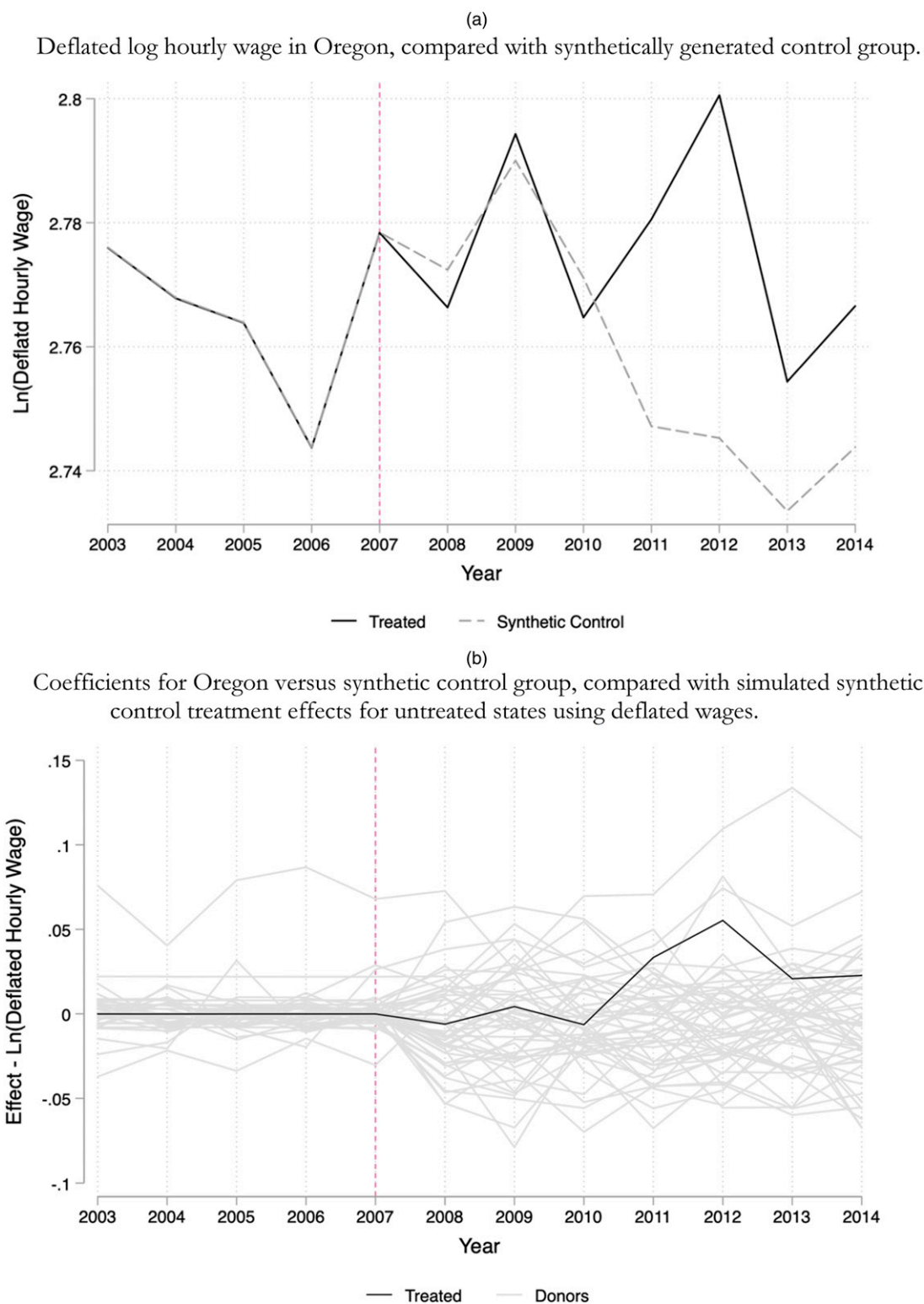
We can quantify this effect using a method suggested by Abadie et al. (2010): we calculate the ratio of the post-treatment RMSPE to the pretreatment RMSPE, and rank the ratio for Oregon in the "placebo" distribution. The rationale behind this method is that the posttreatment RMSPE may be large for two reasons: first, because there is a real treatment effect occurring, which is causing the posttreatment series to deviate from the synthetic control series. Second, the series may simply be relatively noisy. If noise is driving a large posttreatment RMSPE for the treated group, then normalizing by the pretreatment RMSPE causes the rank of the treated group's post- to pretreatment RMSPE ratio to decrease against the distribution of placebo ratios, assuming that post- and pretreatment noise is correlated. In this case, using 2003–2014 as the treatment window, only 4% of placebo observations had a greater ratio of post- to pretreatment RMSPE than Oregon, implying a p -value of 0.04.

Using the weights generated by the synthetic control procedure, we can re-estimate the baseline difference-in-difference model for wages. The results are comparable to those estimated in Table 2: the estimated coefficient is 0.020 ($p < 0.01$), regardless of included fixed effects and controls. Overall, this method bolsters the causal interpretation of our results by suggesting that treatment effects are not being driven by control states that differ greatly from Oregon pre-treatment.

Figure A.1. Synthetic Control Estimates

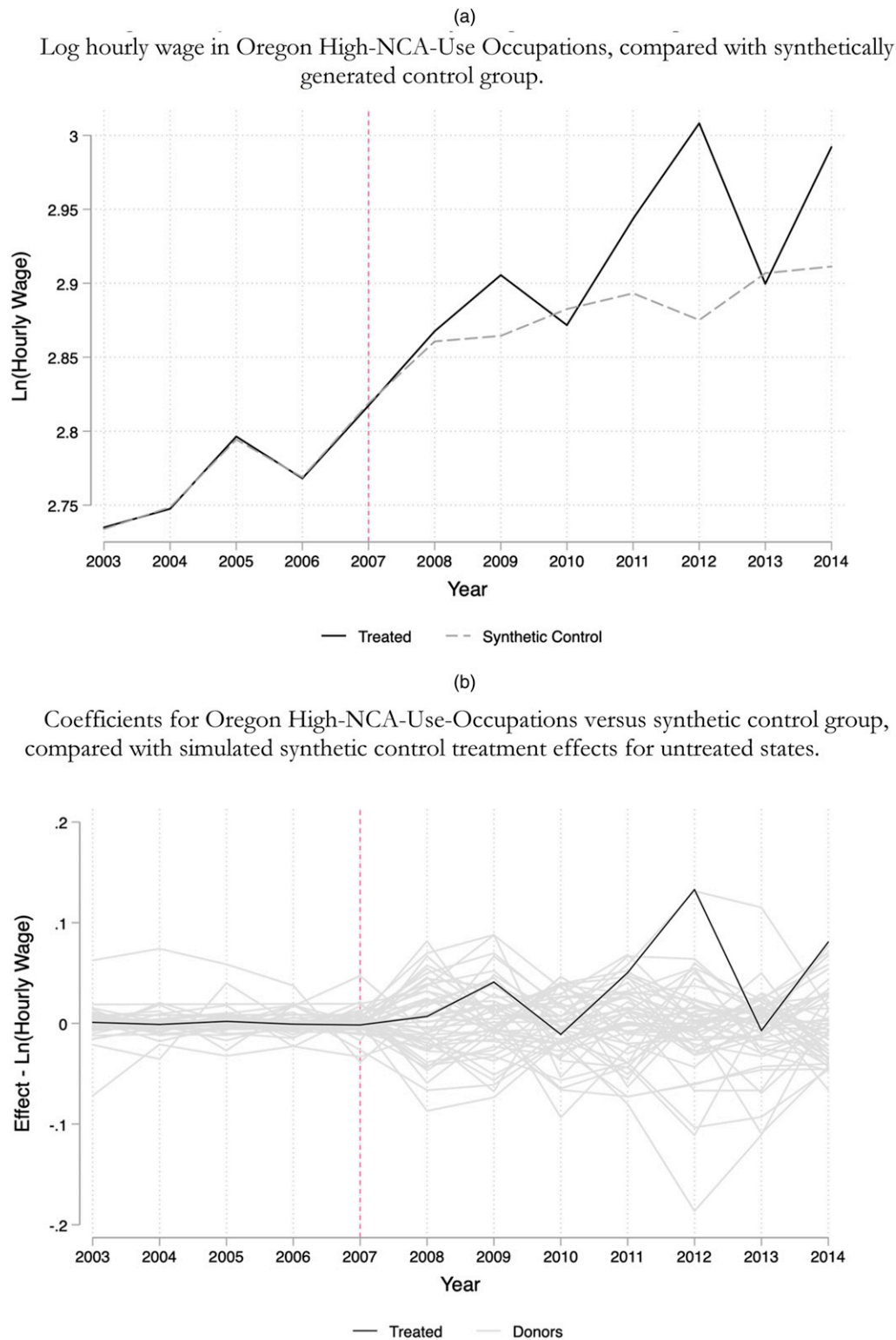


Notes. Panel (a): Log hourly wage in Oregon, compared with synthetically generated control group. Panel (b): Coefficients for Oregon vs. synthetic control group, compared with simulated synthetic control treatment effects for untreated states.

Figure A.2. Synthetic Control Estimates with Deflated Hourly Wage

Notes. Panel (a): Deflated log hourly wage in Oregon, compared with synthetically generated control group. Panel (b): Coefficients for Oregon vs. synthetic control group, compared with simulated synthetic control treatment effects for untreated states using deflated wages.

Figure A.3. Synthetic Control Estimates for High-NCA-Use Occupations



Notes. Panel (a): Log hourly wage in Oregon High-NCA-Use Occupations, compared with synthetically generated control group. Panel (b): Coefficients for Oregon High-NCA-Use Occupations vs. synthetic control group, compared with simulated synthetic control treatment effects for untreated states.

Endnotes

¹ As an example, Figures OA1 and OA2 in the online appendix show the NCA and prohibited regions of a Jimmy John's sandwich maker.

² Statistics related to NCAs among hourly workers are drawn from the 2014 Noncompete Survey Project (for details on the survey, see Starr et al. 2021). We gratefully acknowledge the authors for sharing their data.

³ We note that, whereas the bill rendered NCAs for low-wage workers unenforceable, it did not “ban” them in the strictest sense of the word: there is no codified penalty for an employer who chooses to include an NCA in a low-wage worker's employment contract. However, the law did impose costs on employers for enforcing NCAs through a garden leave provision, which is similar to such a penalty. In any case, with these caveats, we use the word “ban” for brevity.

⁴ For example, Friedman (1991) writes, “The employees who sign non-competition agreements are typically executives or professionals, likely to know at least as much about the likely effect on their future career as their employer.” Similarly, in her discussion of bargaining sophistication, Callahan (1985, pp. 721–722) writes, “An assessment of the kinds of employees likely to enter into express post-employment restraint agreements suggests that the fear that these employees are unable to protect themselves is misplaced. Generally, information worth protecting by contract is, because of its value, not widely dispersed throughout a firm. One can expect that sensitive information would tend to be in the possession of more highly skilled employees. To the extent that such employees can be presumed to be relatively sophisticated and to have alternative employment opportunities, the fear of overreaching by employers diminishes.”

⁵ See Exhibits OA1–OA3 in the online appendix for details of the 2007 Oregon statute (before the ban takes place), as well as the full 2008 bill. A summary is also provided by Duboff and King (2007).

⁶ See ORS 653.295 section (1)(b) and ORS 653.020 section (3) at <https://www.oregonlaws.org/ors/2007/653.020>. Oregon's assistance for employers is at https://www.oregon.gov/boli/TA/Pages/ta_faq_noncompete.aspx. Clarification that the law clearly voided NCAs for hourly workers is provided by the Labor Law Center, which writes, “Such non-compete agreements—even voluntary ones—are illegal when applied to hourly or salaried non-exempt employees under Oregon law.” See <https://www.laborlawcenter.com/education-center/exceptions-to-2008-oregon-non-compete-law/> for more.

⁷ Six senators expressed this view in a request to the Government Accountability Office (Young 2019): “The wide use of non-competes has spread from highly technical fields into less technical and lower wage work, where they might reduce wage and benefit competition among employers and restrict employees' upward mobility—for no good reason.”

⁸ The threshold was based on calculations done for the time period of October 1, 2016, through September 30, 2017, by the Department of Health and Human Services, available at <https://www.acf.hhs.gov/ocs/resource/liheap-im-2016-03-state-median-income-estimates-for-optional-use-in-ffy-2016-and-mandatory-use-in-ffy-2017> (accessed April 22, 2019).

⁹ In fact, the FP p -value in column 5 (which performs a within-Oregon comparison) is exactly zero. Though the FP p -value qualitatively corroborates the cluster-robust p -value, it seems extreme. Accordingly, we re-estimate the p -value using a third method (the wild cluster bootstrap; see Roodman et al. 2019) and find a p -value of 0.033.

¹⁰ For example, the garden leave provision in the 2008 law may have also dramatically reduced the use of NCAs among salaried workers. Or, the early notice provision may have allowed salaried workers to select out of jobs that came with an NCA. Furthermore, salaried workers are far more likely to sign NCAs. Given these changes to the laws and the compositional differences, it is unclear what any

hourly versus salaried comparison would reveal about the impacts of the NCA ban.

¹¹ In addition to the change in the dependent variable, we use lagged values of all control variables to represent the worker's situation in the job from which he or she is leaving. We are therefore forced to omit the union status of the employee, which is only asked in months 4 and 8.

¹² Interestingly, those earning just below the threshold are less likely to be salaried, suggesting that perhaps these workers are being switched to hourly compensation after the passage of Oregon's new NCA law. Another possibility is that salaried workers are given raises in order to push them above the threshold. This would increase the prevalence of salaried work above the threshold and decrease the prevalence below the threshold. Whereas, if correct, this result would be a positive side effect of the NCA ban, it does not explain the increased prevalence of salary compensation in the population at large.

¹³ On June 8, 2014, an article was published in the *New York Times* titled “Noncompete Clauses Increasingly Pop Up in Array of Jobs,” by Steven Greenhouse. This article was soon followed, on October 13, 2014, by an exposé by Dave Jamieson in the Huffington Post on NCAs used by the sandwich shop Jimmy John's, titled “Jimmy John's Makes Low-Wage Workers Sign ‘Oppressive’ Noncompete Agreements.” These two articles, as well as additional ensuing attention from the media and policymakers, likely increased the salience of NCAs in low-wage workers' contracts, on both the employer and worker sides. To avoid any issues that may arise with this increased attention, our preferred estimates are those that end in 2014.

¹⁴ Differences relative to Table 3 stem from the fact that the sample is limited to observations in the CPS that report the county of the household: for respondent confidentiality, county information of many respondents is suppressed.

¹⁵ In this sense, NCAs may operate like no-poach agreements (Krueger and Ashenfelter 2018, Gibson 2019), in that even though workers voluntarily agree to NCAs in theory, in practice they may be invisible to the worker. One alternative theory is that low-wage labor markets are competitive but that low-wage workers develop industry-specific human capital such that within-industry prohibitions (like NCAs) nevertheless provide the firm with labor market power (over the within-industry component of their human capital). Our findings that the positive wage effects from the NCA ban are present for workers of all ages and in many occupations and industries cast doubt on this theory.

¹⁶ Whereas several states did ban NCAs for low-wage workers in 2018 and 2019, the lack of data following those bans make such a study infeasible at this point in time. As more states adopt these bans and more data become available as time passes, such an analysis will be a valuable and feasible way to corroborate the results we document here.

¹⁷ The recommendation in Ferman et al. (2018) is to discard specifications that do not perform matching on a subset or a function of pretreatment outcomes, in order to reduce p -hacking via specification-searching. We note that our estimates are robust to matching on all pretreatment control variables in addition to outcome variables. We also explored a second matching procedure in which we match on outcomes in 2003, 2004, and 2005, and we then ensure that post-treatment trends driving the treatment effect are not present prior to 2008 (in order to test the goodness of the match). In unreported results, we find no evidence of a pretreatment trend when we implement this procedure.

¹⁸ We repeat this analysis in Figure A.2 in the online appendix using deflated hourly wages and in Figure A.3 using high use NCA occupations only.

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